Academy Review Edition · Scalar–Tensor & Disformal Action Mapping · Modules 23–25

Time Attraction Law — Scalar–Tensor Yukawa Fit (V26)

Steps 23–25: GW170817 ($c_T = c$) & Gravitational Lensing, Bullet Cluster (1E 0657-558), and Cosmology & CMB Acoustic Peaks at Fixed $L = 10$ km, $\alpha_Y = 1$

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Modules and Reproducibility Files Included in This Review Document

This document compiles the three extended observational modules developed on top of the V25 curved-exterior solution (10.5281/zenodo.23699685) at fixed $L = 10$ km and $\alpha_Y = 1$:

- **Step 23 (step23_gw170817_and_lensing_en.md):** Gravitational-wave propagation ($c_T = c$ on the purely conformal minimal branch, $G_{4,X} = 0$, $G_5 = 0$; GW170817 / GRB 170817A), single-action DEF normalization ($\phi_{\rm canon} = \phi_L/\sqrt{2\tilde{\alpha}_L}$, $\alpha_{\rm DEF} = d\ln A/d\phi_{\rm canon} = -1/\sqrt{4\pi}$, $\beta_{\rm DEF} = 0$, $\beta_{\rm Yuk} = \alpha_{\rm DEF}^2 = 1/(4\pi)$, $\omega_{\rm BD} = 2\pi/\alpha_Y^3 - 3/2 \approx 4.7832$), massive Yukawa scalar dispersion ($f_{\rm cut} = 4771.35$ Hz), curved-spacetime light deflection on both the Step 22 1.509998 $M_\odot$ Branch II-Fluid diagnostic SLy neutron-star background (Table 2, $\hat{\alpha}_{\rm Yuk,conf}(b_{\rm graz}) = 1.290852$ rad) and the self-consistent 1.562035 $M_\odot$ Branch II-Conf trace-sourced ($\tilde{\varepsilon} - 3\tilde{P}$) BVP solution (Table 2B, $\hat{\alpha}_{\rm Yuk,conf}(b_{\rm graz}) = 1.334024$ rad), and the dedicated 1.0 $M_\odot$ Solar–Cassini calculation ($\gamma_{\rm PPN,\odot} = 1$, Table 3). Solver script: gw170817_and_lensing_audit.py.
- **Step 24 (step24_bullet_cluster_en.md):** Bullet Cluster (1E 0657-558, $z_L = 0.296$) as a *synthetic 1D illustrative test* ($M_{\rm synth} = 32.089$ vs. 1.637; not an observational χ^2 fit), local $L = 10$ km Yukawa no-go on cluster scales, and the model-dependent 1D King-profile threshold $\eta_{\rm gal}/\eta_{\rm gas} \geq 25.665$. Solver script: bullet_cluster_lensing_audit.py.
- **Step 25 (step25_cmb_and_cosmology_en.md):** Cosmological FLRW background integrals ($H(z)$, $r_s(z_*)$, $D_M(z_*)$, θ_*), conditional separate-branch hypothesis for negative $\rho_\Lambda = -5.96 \times 10^{-27}$ kg/m^3 and compensating $\rho_{\rm clock,cosmo} = +2|\rho_\Lambda|$, external Planck/literature inputs (H_2/H_1, H_3/H_2, input $\Omega_c h^2/\Omega_b h^2 = 5.3643$), negligibility of local $L = 10$ km Yukawa $\Delta G_{\rm eff}/G \sim 10^{-36}$–$10^{-34}$, and the 6-scale grand synthesis table. Solver script: cmb_and_cosmology_audit.py.

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1 Step 23 (Module 1): GW170817 Tensor Wave Speed ($c_T = 1$), Massive Scalar Dispersion, and Gravitational Lensing ($\Phi + \Psi$, $\gamma_{\mathrm{PPN}}(r)$, Curved-Spacetime Light Deflection $\hat{\alpha}(b)$, Shapiro Time Delay, and Weak-Lensing Convergence κ) at $L = 10$ km, $\alpha_Y = 1$

Source file: step23_gw170817_and_lensing_en.md

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Version: V26
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Previous Steps: step1_dilaton_form_en.md – step22_curved_exterior_yukawa_adm_mass_mr_and_coupled_stability_en
(V24 Zenodo DOI: 10.5281/zenodo.23680715; V25 Zenodo DOI: 10.5281/zenodo.23699685)
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Local Sources and Runnable Code:

- Runnable solver code: scalar-tensor-yukawa-fit/gw170817_and_lensing_audit.py (synced to code/gw170817_and_le
- Generated output files: gw170817_dispersion_and_cT.csv, lensing_deflection_and_ppn_slip.csv, gw170817_and_lensing_summary.json (synced to csv/ and data/v23/)

1.1 1. Fundamental Action Principle and Coupling Dictionary ($L = 10$ km, $\alpha_Y = 1$)

In the covariant scalar–tensor formulation of the Time Attraction Law (TAL, step1–step22, V25 DOI: 10.5281/zenodo.23699685), the Einstein-frame gravitational and massive Yukawa scalar action reads:

$$S = \int d^4x\,\sqrt{-g}\left[\frac{c^4}{16\pi G}\,R - \frac{c^4}{16\pi G\,\tilde{\alpha}_L}\left(g^{\mu\nu}\partial_\mu\phi_L\partial_\nu\phi_L + 2\,V(\phi_L)\right)\right] + S_m[\tilde{g}_{\mu\nu},\,\Psi_m]\,,$$

where the massive Yukawa screening potential is:

$$V(\phi_L) = \frac{1}{2}\,m_\phi^2\,\phi_L^2, \qquad m_\phi = \frac{1}{L} = 0.1\ \mathrm{km}^{-1} = 10^{-4}\ \mathrm{m}^{-1},$$

and the coupling dictionary (step1–step5, step14, step22) is:

$$\alpha_Y = 4\pi\,\alpha_{\mathrm{std}} = 8\pi\,\tilde{\alpha}_L = 8\pi\,\beta^2, \qquad \gamma_L = \frac{GL^2}{c^2} = 7.42616027\times10^{-20}\ \mathrm{m^3\,kg^{-1}}.$$

Single action, single field variable (V26 normalization fix). All statements of Steps 23–25 are derived from the one action $S[\phi_L]$ above with $\tilde{\alpha}_L = \alpha_Y/(8\pi)$ and the matter coupling $S_m[A^2(\phi_L)g_{\mu\nu}, \Psi_m]$, $A(\phi_L) = \exp(-\alpha_Y\phi_L)$, $d\ln A/d\phi_L = -\alpha_Y$ ($\phi_L > 0$ near matter, $g^{(J)}_{\mu\nu} = \exp(-2\alpha_Y\phi_L)g_{\mu\nu}$). If the kinetic prefactor inside $\frac{c^4}{16\pi G}[\dots]$ is changed from $1/\tilde{\alpha}_L$ to 2 (Damour–Esposito-Farèse form), the field **must** be rescaled; the same symbol ϕ_L with unchanged $A = \exp(-\alpha_Y\phi_L)$ cannot be kept:

$$\frac{1}{\tilde{\alpha}_L}(\partial\phi_L)^2 = 2(\partial\phi_{\mathrm{canon}})^2 \iff \phi_{\mathrm{canon}} = \frac{\phi_L}{\sqrt{2\tilde{\alpha}_L}} = \sqrt{\frac{4\pi}{\alpha_Y}}\,\phi_L, \qquad \phi_L = \sqrt{\frac{\alpha_Y}{4\pi}}\,\phi_{\mathrm{canon}},$$

$$S_{\mathrm{can}} = \int d^4x\,\sqrt{-g}\,\frac{c^4}{16\pi G}\Big[R - 2\,g^{\mu\nu}\partial_\mu\phi_{\mathrm{canon}}\partial_\nu\phi_{\mathrm{canon}} - 2\,m_\phi^2\phi_{\mathrm{canon}}^2\Big] + S_m\big[A^2 g_{\mu\nu},\,\Psi_m\big]\,, \qquad A = \exp\Big(-\alpha_Y\sqrt{2\tilde{\alpha}_L}\,\phi_{\mathrm{canon}}\Big),$$

$$\frac{d\ln A}{d\phi_{\mathrm{canon}}} = \sqrt{2\tilde{\alpha}_L}\,\frac{d\ln A}{d\phi_L} = -\alpha_Y\sqrt{\frac{\alpha_Y}{4\pi}} = -\frac{1}{\sqrt{4\pi}} = -0.2820947918 \quad (\alpha_Y = 1).$$

Equivalently, with the 4π-normalized variable $\hat{\phi}_{\mathrm{canon}} \equiv \phi_{\mathrm{canon}}/\sqrt{4\pi} = \phi_L/\sqrt{\alpha_Y}$ (kinetic term $-\frac{c^4}{2G}(\partial\hat{\phi}_{\mathrm{canon}})^2$), $d\ln A/d\hat{\phi}_{\mathrm{canon}} = -\alpha_Y^{3/2}$. The fifth-force strength is therefore

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$$\beta_{\rm Yuk} \;=\; \Big(\frac{d\ln A}{d\phi_{\rm canon}}\Big)^2 \;=\; \frac{(d\ln A/d\hat{\phi}_{\rm canon})^2}{4\pi} \;=\; 2\tilde{\alpha}_L\Big(\frac{d\ln A}{d\phi_L}\Big)^2 \;=\; 2\tilde{\alpha}_L\,\alpha_Y^2 \;=\; \frac{\alpha_Y^3}{4\pi} \;=\; \frac{1}{4\pi} \;\approx\; 0.0795774715 \quad (\alpha_Y = 1).$$

Varying $S[\phi_L]$ in the non-relativistic dust / weak-field limit ($\tilde{P} \ll \tilde{\varepsilon} \simeq \rho_m c^2$, $|\phi_L| \ll 1$) gives the linearized static source equation and the Jordan-frame Newtonian potential:

$$\big(\nabla^2 - L^{-2}\big)\phi_L \;=\; -8\pi\tilde{\alpha}_L\,\alpha_Y\,\frac{G\rho_m}{c^2} \;=\; -\alpha_Y^2\,\frac{G\rho_m}{c^2}, \qquad \tilde{\Psi}(r) \;=\; -\frac{GM}{r}\big(1+\beta_{\rm Yuk}e^{-r/L}\big), \qquad \frac{G_{\rm eff}(k)}{G} \;=\; 1+\beta_{\rm Yuk}\frac{k^2L^2}{1+k^2L^2}.$$

At $\alpha_Y = 1$ the value $\beta_{\rm Yuk} = 1/(4\pi)$ is unchanged from V25; for $\alpha_Y \neq 1$ the consistent conformal scaling is $\beta_{\rm Yuk} = 2\tilde{\alpha}_L\alpha_Y^2 = \alpha_Y^3/(4\pi)$ (`BETA_YUK = 2*ALPHA_TILDE_L*ALPHA_Y**2`), with effective Brans–Dicke parameter $\omega_{\rm BD} = \frac{1}{2\beta_{\rm Yuk}} - \frac{3}{2} = \frac{2\pi}{\alpha_Y^3} - \frac{3}{2} = 4.7831853072$ at $\alpha_Y = 1$ (distinguished from the legacy `step5` 4π-convention formula $\frac{1-3\alpha_{\rm std}}{2\alpha_{\rm std}} = \frac{4\pi}{\alpha_Y} - \frac{3}{2} = 11.0663706144$). Here $\alpha_{\rm DEF} \equiv d\ln A/d\phi_{\rm canon} = -1/\sqrt{4\pi} = -0.2820947918$ denotes the first-order Damour–Esposito-Farèse coupling derivative (with second derivative $\beta_{\rm DEF} \equiv d^2\ln A/d\phi_{\rm canon}^2 = 0$), while $\beta_{\rm Yuk} \equiv \alpha_{\rm DEF}^2 = 1/(4\pi)$ is the dimensionless fifth-force ratio.

1.1.1 1.1. Exact Comparison of `Branch II-Conf` (Step 23) vs. `Branch II-Fluid` (Step 22): Interior Equations, Frame Mapping, and Exterior Vacuum Identity

While `Branch II-Fluid` (`step3`, `step4`, `step7`, `step12`, `step22`) and `Branch II-Conf` (`step1`, `step2`, `step5`, `step14`, `step23`–`step25`) share the exact same Einstein-frame gravitational and massive scalar action $S_{g+\phi}[g_{\mu\nu}, \phi_L]$, they use two distinct interior matter couplings for a barotropic fluid with equation of state $\tilde{\varepsilon}(h), \tilde{P}(h), \tilde{\rho}_0(h)$ ($dh = d\tilde{P}/(\tilde{\varepsilon}+\tilde{P})$):

1. **`Branch II-Fluid` (`yukawa_action_el` / Step 22 direct fluid-density coupling):**
   - *Matter Lagrangian density:* $\mathcal{L}_m^{(\text{II-Fluid})} = -e^{-\phi_L}\,\varepsilon(n)$, with $P(n) = n\,d\varepsilon/dn - \varepsilon(n)$ and Einstein-frame fluid stress-energy tensor $T^{\mu\nu}_{\text{matter,(II-Fluid)}} = e^{-\phi_L}[(\varepsilon+P)u^\mu u^\nu + Pg^{\mu\nu}]$, so $\rho_m^{(E)} = e^{-\phi_L}\varepsilon(h)$ and $P_m^{(E)} = e^{-\phi_L}P(h)$.
   - *Einstein (m', ν'), scalar (ϕ_L''), and Euler–TOV (h') equations ($B(r) = 1 - 2m(r)/r$, $G = c = 1$):*

$$\frac{dm}{dr} = 4\pi r^2\big(e^{-\phi_L}\varepsilon + \rho_\phi\big), \qquad \frac{d\nu}{dr} = \frac{m + 4\pi r^3\big(e^{-\phi_L}P + p_{r,\phi}\big)}{r(r-2m)}, \qquad \frac{dh}{dr} = -\nu'(r) + \phi_L'(r),$$

$$B\,\phi_L'' + \left(\frac{2B}{r} + B\nu' + \frac{B'}{2}\right)\phi_L' - \frac{\phi_L}{L^2} \;=\; -\frac{\alpha_Y G}{c^2}\,e^{-\phi_L}\,\varepsilon(r), \qquad \rho_\phi = \frac{B(\phi_L')^2 + \phi_L^2/L^2}{2\alpha_Y}, \quad p_{r,\phi} = \frac{B(\phi_L')^2 - \phi_L^2/L^2}{2\alpha_Y}.$$

2. **`Branch II-Conf` (Step 23 universal conformal Jordan coupling):**
   - *Matter action and frame mapping:* $S_m^{(\text{II-Conf})}[\tilde{g}_{\mu\nu}, \Psi_m]$ with $\tilde{g}_{\mu\nu} = A^2(\phi_L)g_{\mu\nu} = e^{-2\alpha_Y\phi_L}g_{\mu\nu}$ and $\sqrt{-\tilde{g}} = A^4(\phi_L)\sqrt{-g} = e^{-4\alpha_Y\phi_L}\sqrt{-g}$. A fluid with proper Jordan-frame energy density $\tilde{\varepsilon}(h)$, pressure $\tilde{P}(h)$, and 4-velocity $\tilde{u}^\mu = A^{-1}u^\mu$ has Jordan stress-energy $\tilde{T}^{\mu\nu} = (\tilde{\varepsilon}+\tilde{P})\tilde{u}^\mu\tilde{u}^\nu + \tilde{P}\tilde{g}^{\mu\nu}$ (trace $\tilde{T} = \tilde{g}_{\mu\nu}\tilde{T}^{\mu\nu} = -\tilde{\varepsilon} + 3\tilde{P}$) and Einstein-frame stress-energy tensor $T^{\mu\nu}_{\text{matter,(II-Conf)}} = \frac{2}{\sqrt{-g}}\frac{\delta S_m}{\delta g_{\mu\nu}} = A^6(\phi_L)\tilde{T}^{\mu\nu}$, i.e., $T^{\mu}{}_{\nu,\text{matter,(II-Conf)}} = e^{-4\alpha_Y\phi_L}\,\mathrm{diag}(-\tilde{\varepsilon}, \tilde{P}, \tilde{P}, \tilde{P})$, so $\rho_m^{(E)} = e^{-4\alpha_Y\phi_L}\tilde{\varepsilon}(h)$, $P_m^{(E)} = e^{-4\alpha_Y\phi_L}\tilde{P}(h)$, and $T^{(E)}_{\text{matter}} = -e^{-4\alpha_Y\phi_L}(\tilde{\varepsilon} - 3\tilde{P})$.
   - *Einstein (m', ν'), scalar (ϕ_L''), and Euler–TOV (h') equations:*

$$\frac{dm}{dr} = 4\pi r^2\big(e^{-4\alpha_Y\phi_L}\tilde{\varepsilon} + \rho_\phi\big), \qquad \frac{d\nu}{dr} = \frac{m + 4\pi r^3\big(e^{-4\alpha_Y\phi_L}\tilde{P} + p_{r,\phi}\big)}{r(r-2m)}, \qquad \frac{dh}{dr} = -\nu'(r) + \alpha_Y\phi_L'(r),$$

$$B\,\phi_L'' + \left(\frac{2B}{r} + B\nu' + \frac{B'}{2}\right)\phi_L' - \frac{\phi_L}{L^2} \;=\; -\frac{\alpha_Y^2 G}{c^2}\,e^{-4\alpha_Y\phi_L}\big(\tilde{\varepsilon}(r) - 3\tilde{P}(r)\big).$$

Structural consequences for Steps 22–25:

- **Exterior vacuum ($r > R_*$, where $\tilde{\varepsilon} = \tilde{P} = 0$):** Both branches satisfy the **exact same 4-component curved exterior Einstein–scalar vacuum ODE system** (`step22`), differing only in the surface boundary values $(R_*, M_{\rm surf}, \phi_L(R_*))$ supplied by the interior integration.
- **Weak-field non-relativistic dust limit ($\tilde{P} \ll \tilde{\varepsilon} \simeq \rho_m c^2$, $|\phi_L| \ll 1$):** At the baseline $\alpha_Y = 1$ ($\alpha_Y^2 = \alpha_Y = 1$), the two linearized sources coincide; for $\alpha_Y \neq 1$, the `Branch II-Conf` source scales as α_Y^2 whereas the `Branch II-Fluid` source scales as α_Y.

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- **Relativistic neutron-star interior ($0 \le r \le R_*$):** Because relativistic pressure reduces the trace ($3\tilde{P}_c/\tilde{\varepsilon}_c = 0.566065$, so $1 - 3\tilde{P}_c/\tilde{\varepsilon}_c = 0.433935$ at the center of the SLy $\log_{10}\rho_{0c} = 15.0$ star) and the conformal volume weight is $e^{-4\alpha_Y\phi_L}$ rather than $e^{-\phi_L}$, the two interior BVPs are **distinct**. Accordingly, in Section 3.3 below, the ray-tracing table on the V25 Step 22 background (`Table 2`, $M_{\rm ADM} = 1.50999814\ M_\odot$) is explicitly labeled as a **conditional cross-branch diagnostic evaluated on the `Branch II-Fluid` background**, and is accompanied by the **self-consistent `Branch II-Conf` trace-sourced interior + exterior BVP solution** (`Table 2B`, $M_{\rm ADM}^{\rm (II\text{-}Conf)} = 1.56203491\ M_\odot$).

At the fixed baseline parameters $L = 10$ km, $\alpha_Y = 1$, the numerical values of the coupling dictionary are:

Quantity	Formula	Numerical Value ($L = 10$ km, $\alpha_Y = 1$)	Unit
Yukawa screening length (L)	$L = 1/m_\phi$	`10.0` (`1.0e4`)	km (m)
Yukawa inverse length (m_ϕ)	$m_\phi = 1/L$	`0.1` (`1.0e-4`)	km^{-1} (m^{-1})
Scalar Compton mass ($m_\phi\hbar c$)	$\hbar c/L$	`1.9732698e-11`	eV
Scalar cutoff angular frequency ($\omega_{\rm cut}$)	c/L	`29979.2458`	$\mathrm{rad\,s}^{-1}$
Scalar cutoff linear frequency ($f_{\rm cut}$)	$c/(2\pi L)$	`4771.345159`	Hz
Yukawa source strength (α_Y)	$4\pi\alpha_{\rm std} = 8\pi\tilde{\alpha}_L$	`1.0`	dimensionless
Standard fifth-force coupling ($\alpha_{\rm std}$)	$\alpha_Y/(4\pi)$	`0.0795774715`	dimensionless
Action normalization constant ($\tilde{\alpha}_L$)	$\alpha_Y/(8\pi)$	`0.0397887358`	dimensionless
Canonical rescaling ($\phi_{\rm canon}/\phi_L$)	$1/\sqrt{2\tilde{\alpha}_L} = \sqrt{4\pi/\alpha_Y}$	`3.5449077018`	dimensionless
First-order DEF derivative ($\lvert\alpha_{\rm DEF}\rvert = \lvert d\ln A/d\phi_{\rm canon}\rvert$; $\beta_{\rm DEF} = d^2\ln A/d\phi_{\rm canon}^2 = 0$)	$\alpha_Y\sqrt{2\tilde{\alpha}_L} = 1/\sqrt{4\pi}$	`0.2820947918`	dimensionless
Fifth-force strength ($\beta_{\rm Yuk}$)	$(d\ln A/d\phi_{\rm canon})^2 = (d\ln A/d\hat{\phi}_{\rm canon})^2/(4\pi) = 2\tilde{\alpha}_L\alpha_Y^2$	`0.0795774715`	dimensionless
Effective Brans–Dicke parameter ($\omega_{\rm BD}$)	$1/(2\beta_{\rm Yuk}) - 3/2 = 2\pi/\alpha_Y^3 - 3/2$	`4.7831853072`	dimensionless

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1.2 2. Part A: GW170817 / GRB 170817A and Tensor Gravitational-Wave Speed ($c_T = 1$)

1.2.1 2.1. Covariant Perturbation Proof in Both Frames (Einstein & Jordan / Horndeski Mapping)

1. **Einstein-Frame Tensor Modes ($g_{\mu\nu} = \bar{g}_{\mu\nu} + h_{\mu\nu}$):** In the Einstein frame, the gravitational kinetic term is purely the Einstein–Hilbert term $\frac{c^4}{16\pi G}R$; the scalar field ϕ_L has no derivative couplings to the Riemann, Ricci, or Einstein tensors. In transverse-traceless (TT) gauge ($h^\mu{}_\mu = 0$, $\bar{\nabla}^\mu h^{\rm TT}_{\mu\nu} = 0$), the quadratic tensor action reads:

$$S_T^{(2)} \;=\; \frac{c^4}{64\pi G}\int d^4x\,\sqrt{-\bar{g}}\,\left[-\bar{g}^{\mu\nu}\partial_\mu h^{\rm TT}_{ij}\,\partial_\nu h^{ij\,\rm TT}\right] \quad\Longrightarrow\quad \Box_{\bar{g}}\,h^{\rm TT}_{ij} \;=\; 0.$$

   Hence the propagation speed of tensor gravitational waves in the Einstein frame is identically:

$$c^2_{T,\rm Einstein} \;=\; c^2 \quad\Longleftrightarrow\quad \frac{c_{T,\rm Einstein} - c}{c} \;=\; 0.$$

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2. **Jordan-Frame Horndeski Functions ($\tilde{g}_{\mu\nu} = A^2(\phi_L) g_{\mu\nu}$):** Transforming to the physical Jordan-frame metric $\tilde{g}_{\mu\nu} = A^2(\phi_L) g_{\mu\nu}$ (in the locked V26 convention $A(\phi_L) = \exp(-\alpha_Y \phi_L)$, $\phi_L > 0$), the scalar–tensor action maps into the Horndeski class with curvature coupling functions:

$$\tilde{G}_4(\phi_L) \;=\; \frac{c^4}{16\pi G}\, A^{-2}(\phi_L), \qquad \tilde{G}_{4,X} \;\equiv\; \frac{\partial \tilde{G}_4}{\partial X} \;=\; 0, \qquad \tilde{G}_5(\phi_L, X) \;=\; 0,$$

where $X \equiv -\frac{1}{2}\tilde{g}^{\mu\nu}\partial_\mu\phi_L\partial_\nu\phi_L$. Because $\tilde{G}_4$ is a purely algebraic function of ϕ_L and independent of the kinetic invariant X ($\tilde{G}_{4,X} = 0$, $\tilde{G}_5 = 0$), the general Horndeski tensor speed formula yields:

$$\boxed{c^2_{T,\mathrm{Jordan}} \;=\; \frac{\tilde{G}_4}{\tilde{G}_4 - 2X\tilde{G}_{4,X}}\, c^2 \;=\; c^2 \quad\Longrightarrow\quad \frac{c_T - c}{c} \;=\; 0 \quad \text{identically for all } (L,\, \alpha_Y)!}$$

Scope (V26): $c_T = c$ is proven on the purely conformal, minimal Einstein–Hilbert + canonical scalar + universal Maxwell coupling branch ($G_{4,X} = 0$, $G_5 = 0$); it does not automatically prove $c_T = c$ on a disformal/refractive branch ($D(\phi_L, X) \neq 0$).

3. **Conformal Invariance of Null Cones:** The conformal transformation $\tilde{g}_{\mu\nu} = A^2(\phi_L) g_{\mu\nu}$ preserves null cones identically ($\tilde{g}_{\mu\nu}k^\mu k^\nu = A^2(\phi_L) g_{\mu\nu}k^\mu k^\nu = 0 \iff g_{\mu\nu}k^\mu k^\nu = 0$). Because four-dimensional vacuum Maxwell electrodynamics ($\sqrt{-\tilde{g}}\,\tilde{g}^{\mu\alpha}\tilde{g}^{\nu\beta}F_{\mu\nu}F_{\alpha\beta} = \sqrt{-g}\, g^{\mu\alpha}g^{\nu\beta}F_{\mu\nu}F_{\alpha\beta}$) is conformally invariant, electromagnetic waves (photons) and tensor gravitational waves $h^{\mathrm{TT}}_{+,\times}$ propagate along the exact same null cone.

   - **Comparison with the GW170817 / GRB 170817A Observational Bound** ($\Delta t = 1.74 \pm 0.05$ s over $D = 40$ Mpc $= 1.2343 \times 10^{20}$ km, light travel time $t_{\mathrm{light}} = 4.1171 \times 10^{15}$ s $\approx 1.3046 \times 10^8$ yr, Abbott et al. 2017):

$$-3 \times 10^{-15} \;\le\; \frac{c_T - c}{c} \;\le\; +7 \times 10^{-15}, \qquad \left(\frac{c_T - c}{c}\right)_{\mathrm{TAL}} \;=\; 0.$$

     Both symbolic (`SymPy`) and numerical checks (`gw170817_and_lensing_summary.json`) confirm a residual of `0` in both frames.

1.2.2 2.2. Massive Scalar Breathing Mode Dispersion and the 4.77 kHz Cutoff

Linearizing the vacuum scalar wave equation ($\Box\phi_L - m_\phi^2\phi_L = 0$, $m_\phi = 1/L$) for plane waves ($\delta\phi_L \propto e^{i(\mathbf{k}\cdot\mathbf{x} - \omega t)}$) yields the Klein–Gordon dispersion relation:

$$\omega^2 \;=\; c^2k^2 + \omega_{\mathrm{cut}}^2, \qquad \omega_{\mathrm{cut}} \;=\; \frac{c}{L} \;=\; 29\,979.2458\ \mathrm{rad\,s^{-1}}, \qquad f_{\mathrm{cut}} \;=\; \frac{c}{2\pi L} \;=\; 4771.345159\ \mathrm{Hz}.$$

1. **Sub-Cutoff (Evanescent / Yukawa-Screened) Regime ($f < f_{\mathrm{cut}} = 4771.345$ Hz):** At all frequencies below the 4.771 kHz Compton cutoff — including binary-pulsar orbital frequencies ($f \sim 10^{-4}$ Hz), the LISA band (10^{-2} Hz), the entire GW170817 LIGO inspiral band ($f_{\mathrm{GW}} \in [24, 600]$ Hz), and even the post-merger neutron-star fundamental quadrupole f-mode ($f \sim 1500$–3000 Hz) — the wavenumber is purely imaginary:

$$k \;=\; i\,\kappa_{\mathrm{ev}}(f), \qquad \kappa_{\mathrm{ev}}(f) \;=\; \frac{1}{L}\sqrt{1 - \left(\frac{f}{f_{\mathrm{cut}}}\right)^2}, \qquad \lambda_{\mathrm{ev}}(f) \;=\; \frac{1}{\kappa_{\mathrm{ev}}(f)} \;=\; \frac{L}{\sqrt{1 - (f/f_{\mathrm{cut}})^2}} \;\approx\; 10\ \mathrm{km}.$$

   The scalar breathing mode does not propagate as a wave; it is an evanescent near-zone field decaying exponentially over $\lambda_{\mathrm{ev}}(f) \approx 10$–$12.86$ km. Over the GW170817 propagation distance $D = 40$ Mpc $= 1.2343 \times 10^{20}$ km, the scalar amplitude suppression is:

$$\exp\!\left(-\frac{D}{\lambda_{\mathrm{ev}}(f)}\right) \;\sim\; 10^{-5.36\times 10^{19}} \;=\; 0.$$

2. **Super-Cutoff (Propagating Dispersive) Regime ($f > f_{\mathrm{cut}} = 4771.345$ Hz):** If a high-frequency post-merger burst excited modes with $f > 4771.345$ Hz, their group and phase velocities would be:

$$v_g(f) \;=\; \frac{\partial\omega}{\partial k} \;=\; c\sqrt{1 - \left(\frac{f_{\mathrm{cut}}}{f}\right)^2} \;<\; c, \qquad v_p(f) \;=\; \frac{\omega}{k} \;=\; \frac{c^2}{v_g(f)} \;>\; c.$$

   Over $D = 40$ Mpc, the arrival time delay of a super-cutoff scalar mode relative to the $c_T = c$ tensor wave is:

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- At $f = 5000$ Hz ($v_g/c = 0.298949$): $\Delta t_{\rm scalar} = 9.655 \times 10^{15}$ s $= 3.059 \times 10^8$ yr (305.9 million years),
- At $f = 10\,000$ Hz ($v_g/c = 0.878830$): $\Delta t_{\rm scalar} = 5.676 \times 10^{14}$ s $= 1.799 \times 10^7$ yr (18.0 million years). Consequently, terrestrial detectors at $D = 40$ Mpc observe purely the $c_T = c$ tensor polarizations $h_+, h_\times$ with zero scalar contamination.

Table 1: GW170817 Tensor Wave Speed and Massive Scalar Mode Dispersion ($L = 10$ km, $\alpha_Y = 1$; `gw170817_dispersion_and_cT.csv`)

Mode / Band (`mode_label`)	f (Hz)	$f/f_{\rm cut}$	Scalar Regime	c_T/c (Einstein & Jordan)	$(c_T - c)/c$	$k_{\rm real}$ (km^{-1})	$\kappa_{\rm ev}$ (km^{-1})	$\lambda_{\rm ev}$ (km)	$\log_{10}(A_{\rm scalar}/A_{\rm tensor}(40\ {\rm Mpc}))$	v_g/c	$\Delta t_{\rm scalar}(40$ Mpc) (yr)
PSR_B1913+16_orbital	1.587e-9	3.509e-9	evanescent_Yukawa_screened	1.0	0.0	0.100000	10.000000	-5.360e+19	0.0	0.0	
PSR_J0737-3039_orbital	1.387e-8	3.068e-8	evanescent_Yukawa_screened	1.0	0.0	0.100000	10.000000	-5.360e+19	0.0	0.0	
LISA_mHz_band	1.000e-2	2.096e-6	evanescent_Yukawa_screened	1.0	0.0	0.100000	10.000000	-5.360e+19	0.0	0.0	
GW170817_inspiral_entry	23.000	0.024830	evanescent_Yukawa_screened	1.0	0.0	0.099999	10.000127	-5.360e+19	0.0	0.0	
GW170817_100Hz	100.000	0.020958	evanescent_Yukawa_screened	1.0	0.0	0.099978	10.002197	-5.359e+19	0.0	0.0	
GW170817_300Hz	300.000	0.062875	evanescent_Yukawa_screened	1.0	0.0	0.099802	10.019825	-5.350e+19	0.0	0.0	
GW170817_600Hz	600.000	0.125751	evanescent_Yukawa_screened	1.0	0.0	0.099206	10.080016	-5.318e+19	0.0	0.0	
NS_post_merger_1500Hz	1500.000	0.314377	evanescent_Yukawa_screened	1.0	0.0	0.094930	10.534097	-5.089e+19	0.0	0.0	
NS_f_mode_2500Hz	2500.000	0.523961	evanescent_Yukawa_screened	1.0	0.0	0.085174	11.740643	-4.566e+19	0.0	0.0	
NS_high_3000Hz	3000.000	0.628754	evanescent_Yukawa_screened	1.0	0.0	0.077760	12.860005	-4.168e+19	0.0	0.0	
Sub_cutoff_4500Hz	4500.000	0.943130	evanescent_Yukawa_screened	1.0	0.0	0.033242	30.082119	-1.782e+19	0.0	0.0	
Yukawa_cutoff_4771.345159	4771.345159	1.000000	cutoff_threshold	1.0	0.0	0.000000	inf	0.0	0.0	inf	
Super_cutoff_4800Hz	4800.000	1.006006	propagating_dispersive	0.010976	0.0	0.0	0.0	0.109105	1.065291e+09		
Super_cutoff_5000Hz	5000.000	1.047923	propagating_dispersive	0.031328	0.0	0.0	0.0	0.298949	3.059412e+08		
Super_cutoff_7500Hz	7500.000	1.571884	propagating_dispersive	0.121277	0.0	0.0	0.0	0.771541	3.863088e+07		
Super_cutoff_10000Hz	10000.000	2.095845	propagating_dispersive	0.184189	0.0	0.0	0.0	0.878830	1.798767e+07		

1.3 3. Part B: Gravitational Lensing ($\Phi + \Psi$, $\gamma_{\rm PPN}(r)$, Curved-Spacetime $\hat{\alpha}(b)$, and Convergence κ) at $L = 10$ km, $\alpha_Y = 1$

1.3.1 3.1. Weak-Field Potentials in the Einstein and Jordan Frames: Conformal Cancellation in the Weyl Potential

In Newtonian gauge, the weak-field Einstein-frame metric reads:

$$ds^2 \;=\; -\left(1 + \frac{2\Psi_E(r)}{c^2}\right) c^2 dt^2 \;+\; \left(1 - \frac{2\Phi_E(r)}{c^2}\right) d\mathbf{x}^2, \qquad \Psi_E(r) \;=\; \Phi_E(r) \;=\; -\frac{GM_{\rm ADM}}{r}.$$

In the physical Jordan-frame metric ($\tilde{g}_{\mu\nu} = A^2(\phi_L) g_{\mu\nu} = \exp(-2\alpha_Y \phi_L) g_{\mu\nu}$, $A(\phi_L) = \exp(-\alpha_Y \phi_L)$, $\phi_L > 0$), the temporal ($\tilde{\Psi}$) and spatial ($\tilde{\Phi}$) metric potentials are shifted at linear order with opposite signs:

$$\tilde{\Psi}(r) \;=\; \Psi_E(r) - \alpha_Y c^2 \phi_L(r), \qquad \tilde{\Phi}(r) \;=\; \Phi_E(r) + \alpha_Y c^2 \phi_L(r).$$

Consequently, **in the Weyl lensing potential governing null geodesics (gravitational lensing and Shapiro time delay at fixed $M_{\rm ADM}$), the conformal factor cancels identically at linear order**:

$$\Phi_{\rm lens}(r) \;\equiv\; \frac{1}{2}\big(\tilde{\Psi}(r) + \tilde{\Phi}(r)\big) \;=\; \frac{1}{2}\big(\Psi_E(r) + \Phi_E(r)\big) \;=\; -\frac{GM_{\rm ADM}}{r} \quad \text{(outside the Yukawa energy cloud).}$$

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1.3.2 3.2. Scale-Dependent PPN Slip ($\gamma_{\rm PPN}(r)$) and the Lensing-to-Dynamical Mass Ratio

Whereas relativistic photons respond to the Weyl combination $\Phi_{\rm lens}(r) = \frac{1}{2}(\tilde{\Psi} + \tilde{\Phi}) = -GM_{\rm ADM}/r$, non-relativistic test bodies (planets, stars, gas clouds) respond to the dynamical Newtonian potential:

$$\tilde{\Psi}(r) \;=\; -\frac{GM_{\rm ADM}}{r}\big(1 + \beta_{\rm Yuk}\, e^{-r/L}\big), \qquad \tilde{\Phi}(r) \;=\; -\frac{GM_{\rm ADM}}{r}\big(1 - \beta_{\rm Yuk}\, e^{-r/L}\big).$$

This yields the scale-dependent Eddington–Robertson PPN slip parameter $\gamma_{\rm PPN}(r) \equiv \tilde{\Phi}(r)/\tilde{\Psi}(r)$ and the lensing-to-dynamical potential ratio:

$$\boxed{\gamma_{\rm PPN}(r) \;=\; \frac{1 - \beta_{\rm Yuk}\, e^{-r/L}}{1 + \beta_{\rm Yuk}\, e^{-r/L}}, \qquad \frac{\Phi_{\rm lens}(r)}{\tilde{\Psi}(r)} \;=\; \frac{1 + \gamma_{\rm PPN}(r)}{2} \;=\; \frac{1}{1 + \beta_{\rm Yuk}\, e^{-r/L}}.}$$

The V26 package uses a single normalization (`lensing_deflection_and_ppn_slip.csv`): $\beta_{\rm Yuk}(\alpha_Y) = 2\tilde{\alpha}_L \alpha_Y^2 = \alpha_Y^3/(4\pi) = 1/(4\pi) \approx 0.0795774715$ at $\alpha_Y = 1$ ($\tilde{\alpha}_L = \frac{1}{8\pi}$). No parallel $\beta_{\rm Yuk} = 2.0$ column is reported.

At the Cassini 2003 solar conjunction impact parameter ($b = 1.6\, R_\odot = 1.11312 \times 10^6$ km, $b/L = 1.11312 \times 10^5$), the exponential Yukawa factor is $e^{-b/L} = 10^{-48\,342.19} = 0$, giving:

$$\gamma_{\rm PPN}(1.6\, R_\odot) \;=\; 1, \qquad \big|1 - \gamma_{\rm PPN}(1.6\, R_\odot)\big| \;<\; 10^{-48\,342} \;\ll\; 2.3 \times 10^{-5} \text{ (Bertotti et al. 2003).}$$

1.3.3 3.3. Exact Curved-Spacetime Deflection Angle ($\hat{\alpha}(b)$) and Shapiro Time Delay: Conditional Diagnostic on the V25 Step 22 (`Branch II-Fluid`) Background vs. Self-Consistent `Branch II-Conf` Trace-Sourced Solution

As established in Section 1.1, the V25 Step 22 neutron-star background (`curved_yukawa_adm_and_stability.py`) integrates the interior with the `Branch II-Fluid` coupling $\mathcal{L}_m = -e^{-\phi_L}\varepsilon(n)$ (source $-\alpha_Y e^{-\phi_L}\varepsilon$), whereas `Branch II-Conf` sources ϕ_L inside a relativistic star via the trace $-\alpha_Y^2 e^{-4\alpha_Y\phi_L}(\tilde{\varepsilon} - 3\tilde{P})$. Evaluating null-geodesic light deflection on the V25 Step 22 metric is therefore a **conditional cross-branch diagnostic on the `Branch II-Fluid` background**, which we present in **Table 2** alongside the **self-consistent `Branch II-Conf` trace-sourced interior + exterior BVP solution** in **Table 2B** (`gw170817_and_lensing_summary.json`, key `branch_ii_fluid_vs_branch_ii_conf_interior_bvp_comparison`).

For the V25 (`step22`, DOI: `10.5281/zenodo.23699685`, `Branch II-Fluid` background) reference SLy neutron star ($\log_{10}(\rho_{0c}/[\text{g cm}^{-3}]) = 15.0$, $L = 10$ km, $\alpha_Y = 1$), the exact 4-component exterior BVP parameters are:

- Stellar radius: $R_* = 11.46572423$ km,
- Coordinate-radius surface mass: $M_{\rm surf} = m(R_*) = 1.50766122\, M_\odot$ (2.22625027 km),
- Asymptotically flat ADM mass: $M_{\rm ADM} = \lim_{r\to\infty} m(r) = 1.50999814\, M_\odot$ (2.22970113 km),
- Exterior scalar cloud mass contribution: $\Delta M^{\rm curved}_{\phi,\rm ext} = +0.00233692\, M_\odot$ (+0.155003% relative to $M_{\rm surf}$),
- Surface scalar field and curved derivative: $\phi_L(R_*) = 0.0054086823$, $\phi'_{L,\rm curved}(R_*) = -0.0012631563$ km^{-1} ($k_{\rm curved} = 0.23354233$ km^{-1}),
- Surface metric potential: $\nu_{\rm exact}(R_*) = -0.2463066872$ (compared with the vacuum Schwarzschild values $\nu_{\rm Schw}(M_{\rm surf}) = -0.2457823129$ and $\nu_{\rm Schw}(M_{\rm ADM}) = -0.2462745914$).

In the spherically symmetric exterior spacetime ($ds^2 = -e^{2\nu(r)}c^2dt^2 + B(r)^{-1}dr^2 + r^2 d\Omega^2$, $B(r) = 1 - 2m(r)/r$), the exact curved-spacetime deflection angle for a light ray with impact parameter $b = r_0\, e^{-\nu_{\rm eff}(r_0)}$ (evaluated via cancellation-free 128-point Gauss–Legendre quadrature with $u = r_0/r = \sin\theta$, $\theta \in [0, \pi/2]$) is:

$$\hat{\alpha}(b) \;=\; 2\int_0^{\pi/2} \left[\frac{\cos\theta}{\sqrt{B(r_0/\sin\theta)}\sqrt{\exp\big(2[\nu_{\rm eff}(r_0) - \nu_{\rm eff}(r_0/\sin\theta)]\big) - \sin^2\theta}} \;-\; 1\right] d\theta,$$

where on the conformal Jordan branch $\nu_{\rm eff}(r) = \nu(r)$ (since $d\tilde{s}^2 = e^{-2\alpha_Y\phi_L}ds^2 = 0 \iff ds^2 = 0$), while on the disformal refractive branch (`step2`, `step14`, $n_{\rm disf}(r) = e^{\phi_L(r)}$) $\nu_{\rm eff}(r) = \nu(r) - \phi_L(r)$.

Three Physical Consequences from the Stellar Surface ($b = 14.749$ km) to Galaxy Clusters ($b = 1$ Mpc):

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1. **Birkhoff-Violating Effect of the Exterior Yukawa Energy Cloud near the Neutron Star ($b \in [14.75, 100]$ km):**
   - For the surface-grazing ray ($b = 14.749032$ km, closest approach $r_0 = 11.558475$ km), the enclosed mass is $m(r_0) = 1.50772597\ M_\odot < M_{\rm ADM}$.
   - The conformal Yukawa deflection angle is $\hat\alpha_{\rm Yukawa,conf} = 1.29085207$ rad (73.96037°), which is +0.36521% **larger** than the vacuum Schwarzschild deflection with the surface mass ($\hat\alpha_{\rm Schw}(M_{\rm surf}) = 1.28615489$ rad), but −0.02867% **smaller** than the vacuum Schwarzschild deflection with the asymptotic ADM mass ($\hat\alpha_{\rm Schw}(M_{\rm ADM}) = 1.29122231$ rad) because the light ray dives inside the exterior scalar energy halo!
   - On the disformal refractive branch ($n_{\rm disf} = e^{\phi_L}$), the grazing deflection angle is $\hat\alpha_{\rm Yukawa,disf} = 1.34618158$ rad (+4.28628% above the conformal branch), decaying exponentially to $+5.253 \times 10^{-6}$ at $b = 100$ km and 0 at $b \geq 1000$ km.
2. **Asymptotic ADM Mass Calibration ($b \geq 200$ km):**
   - For $b \geq 200$ km $= 20\,L$, the scalar field vanishes ($\phi_L(200\text{ km}) = 1.069\times10^{-12}$), so $\hat\alpha_{\rm Yukawa,conf}(b)$ coincides with $\hat\alpha_{\rm Schw}(M_{\rm ADM}, b)$ to machine precision ($\leq 10^{-12}$ relative difference, and 0.0 for $b \geq R_\odot$), while maintaining a permanent $+0.155003\% = \Delta M^{\rm curved}_{\phi,\rm ext}/M_{\rm surf}$ excess over the naive surface-mass Schwarzschild curve $\hat\alpha_{\rm Schw}(M_{\rm surf}, b)$ at all asymptotic impact parameters.

Table 2: Scale-Dependent PPN Slip $\gamma_{\rm PPN}(b)$, Curved-Spacetime Light Deflection $\hat\alpha(b)$, and Shapiro Time Delay from $b = 14.75$ km to 1 Mpc (`lensing_deflection_and_ppn_slip.csv`)

Scale (`scale_label`)	b (km)	r_0 (km)	b/L	$\log_{10}(e^{-b/L})$	$m(r_0)$ ($M_\odot$)	$\gamma^{\rm TAL}_{\rm PPN}(b)$	$\lvert 1-\gamma^{\rm TAL}_{\rm PPN}\rvert$	$\hat\alpha_{\rm Yuk,conf}$ (rad)	$\hat\alpha_{\rm Schw}(M_{\rm ADM})$ (rad)	$\hat\alpha_{\rm Schw}(M_{\rm surf})$ (rad)	$\frac{\hat\alpha_{\rm Yuk,conf}-\hat\alpha_{\rm Schw}(M_{\rm ADM})}{\hat\alpha_{\rm Schw}(M_{\rm ADM})}$	$\frac{\hat\alpha_{\rm Yuk,conf}-\hat\alpha_{\rm Schw}(M_{\rm surf})}{\hat\alpha_{\rm Schw}(M_{\rm surf})}$	$\Delta t^{\rm Yuk,conf}_{\rm Shapiro}$ (ms)
NS_surface	14.749032	11.558475	1.547490	−0.64051	1.507726	0.964236	3.576e−2	1.290852	1.291222	1.286154	−2.986e−4	+3.652e−3	20.139048
NS_near	15.000000	11.808443	1.574900	−0.65141	1.507913	0.965107	3.489e−2	1.290768	1.291138	1.286076	−2.986e−4	+3.652e−3	17.501587
NS_Yukawa	20.000000	15.808443	2.099867	−0.86861	1.509517	0.978692	2.131e−2	1.290308	1.290678	1.285616	−2.986e−4	+3.652e−3	7.157454
NS_Yukawa	30.000000	25.808443	3.149801	−1.30291	1.509957	0.992107	7.893e−3	1.289498	1.289868	1.284806	−2.986e−4	+3.652e−3	1.964526
NS_Yukawa	50.000000	45.808443	5.249668	−2.17151	1.509998	0.998928	1.072e−3	1.288176	1.288546	1.283484	−2.986e−4	+3.652e−3	0.066547
NS_Yukawa	100.000000	95.808443	10.499336	−4.34291	1.509998	0.999997	3.226e−6	1.285564	1.285934	1.280872	−2.986e−4	+3.652e−3	0.000258
NS_Yukawa	200.000000	195.808443	20.998671	−8.68591	1.509998	1.000000	3.280e−10	1.280461	1.280831	1.275769	−2.986e−4	+3.652e−3	0.000000
NS_far_1000km	1000.000000	995.808443	104.993357	−43.4291	1.509998	1.000000	5.921e−45	1.250089	1.250459	1.245397	−2.986e−4	+3.652e−3	0.000000
Sirius_B	5844.000000	5839.808443	613.960682	−253.801	1.509998	1.000000	2.513e−255	1.201527	1.201897	1.196835	−2.986e−4	+3.652e−3	0.000000
NS_far_6e7m	60000.000000	59995.808443	6299.579357	−2604.29	1.509998	1.000000	0.0	1.282001e−2	1.282001e−2	1.250018e−2	0.0	+2.500e−1	2.503515
NS_far_1e8m	100000.000000	99995.808443	10499.298928	−4340.29	1.509998	1.000000	0.0	8.012488e−3	8.012488e−3	7.812613e−3	0.0	+2.500e−1	2.503351
NS_far_1e9m	1000000.000000	999995.808443	104992.989284	−43402.9	1.509998	1.000000	0.0	5.961853e−4	5.961853e−4	5.812626e−4	0.0	+2.500e−1	2.503080
Outer_Solar	1495978.707000	1495974.515443	157000.000000	−64900.0	1.509998	1.000000	0.0	5.961852e−4	5.961852e−4	5.812626e−4	0.0	+2.500e−1	2.503078
Subparsec	3.085678e+16	3.085678e+16	3.239956e+12	1.340e+12	1.509998	1.000000	0.0	2.890382e−8	2.890382e−8	2.815914e−8	0.0	+2.500e−1	2.503078
Galactic	3.085678e+19	3.085678e+19	3.239956e+15	1.340e+15	1.509998	1.000000	0.0	2.890382e−8	2.890382e−8	2.815914e−8	0.0	+2.500e−1	2.503078
NGC3198_DM_halo	9.257035e+20	9.257035e+20	9.719787e+16	4.031e+16	1.509998	1.000000	0.0	9.609009e−9	9.609009e−9	9.594130e−9	0.0	+2.500e−1	2.503078
Bullet_Cluster	7.714192e+22	7.714192e+22	8.099267e+18	3.350e+18	1.509998	1.000000	0.0	1.156155e−10	1.156155e−10	1.154366e−10	0.0	+2.500e−1	2.503078
Cluster_scale	3.085678e+22	3.085678e+22	3.239956e+18	1.340e+18	1.509998	1.000000	0.0	2.890382e−9	2.890382e−9	2.815914e−9	0.0	+2.500e−1	2.503078

Note (V26): every row of Table 2 is evaluated on the 1.50999814 $M_\odot$ Step 22 `Branch II-Fluid` neutron-star background (`source_body = NS_SLy_1p51Msun`); the `NS_far_field_b_*` rows are far-field benchmark impact parameters only, not solar data. Table 2B below compares this `Branch II-Fluid` diagnostic background directly against the self-consistent `Branch II-Conf` trace-sourced ($\tilde\varepsilon - 3\tilde P$) BVP solution for the same SLy $\log_{10}\rho_{0c} = 15.0$, $L = 10$ km, $\alpha_Y = 1$ star, and Table 3 gives the dedicated $M = 1.000000\ M_\odot$ Solar–Cassini benchmark.

Table 2B: Direct Comparison of the Step 22 `Branch II-Fluid` Background vs. the Self-Consistent Step 23 `Branch II-Conf` Trace-Sourced Interior + Curved-Exterior BVP (SLy, $\log_{10}\rho_{0c} = 15.0$, $L = 10$ km, $\alpha_Y = 1$)

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Quantity / Observable	`Branch II-Fluid` (Step 22 Background Diagnostic)	`Branch II-Conf` (Self-Consistent Trace-Sourced BVP)	Physical Origin of Difference
Interior matter action	$\mathcal{L}_m = -e^{-\phi_L}\varepsilon(n)$	$S_m[e^{-2\alpha_Y\phi_L}g_{\mu\nu}, \Psi_m]$ $(\sqrt{-\tilde{g}} = e^{-4\alpha_Y\phi_L}\sqrt{-g})$	Direct energy-density weight vs. universal Jordan metric
Interior scalar source RHS	$-(\alpha_Y G/c^2)\, e^{-\phi_L}\varepsilon(r)$	$-(\alpha_Y^2 G/c^2)\, e^{-4\alpha_Y\phi_L}(\tilde{\varepsilon} - 3\tilde{P})$	Relativistic pressure trace reduction $(1 - 3\tilde{P}_c/\tilde{\varepsilon}_c = 0.433935)$
Interior Euler–TOV balance	$h'(r) = -\nu'(r) + \phi_L'(r)$	$h'(r) = -\nu'(r) + \alpha_Y\phi_L'(r)$	Identical form at $\alpha_Y = 1$, evaluated on respective $\phi_L(r)$
Exterior ($r > R_*$) vacuum ODEs	4-component curved Einstein–scalar BVP	Identical 4-component curved Einstein–scalar BVP	Vacuum equations coincide identically ($\tilde{\varepsilon} = \tilde{P} = 0$)
Stellar radius R_* (km)	`11.46572423`	`11.66269842`	Smaller $e^{-4\phi_L}$ weight and trace source shift equilibrium radius
Surface mass $M_{\rm surf} = m(R_*)$ ($M_\odot$)	`1.50766122`	`1.56063550`	Interior mass integral with $e^{-4\alpha_Y\phi_L}\tilde{\varepsilon}$
Asymptotic ADM mass $M_{\rm ADM}$ ($M_\odot$)	`1.50999814`	`1.56203491`	$M_{\rm surf} + \Delta M^{\rm curved}_{\phi,\rm ext}$
Exterior scalar cloud mass $\Delta M^{\rm curved}_{\phi,\rm ext}$ ($M_\odot$)	`+0.00233692` (`+0.155003%`)	`+0.00139941` (`+0.089669%`)	Lower surface scalar field due to $-3\tilde{P}$ in trace source
Central / surface scalar field $\phi_L(0)$ / $\phi_L(R_*)$	`0.01777896` / `0.00540868`	`0.01157688` / `0.00414318`	Trace factor $1 - 3\tilde{P}/\tilde{\varepsilon} < 1$ inside core
Curved surface slope $\phi'_{L,\rm curved}(R_*)$ (km^{-1}) / $k_{\rm curved}$ (km^{-1})	`-0.00126316` / `0.23354233`	`-0.00096508` / `0.23293146`	Curved Dirichlet-to-Neumann exterior matching
Grazing impact parameter $b_{\rm graz}$ (km) / r_0 (km)	`14.749032` / `11.558475`	`15.064914` / `11.736304`	Evaluated at $1.0001\, R_* e^{-(\nu(R_*)-\phi_L(R_*))}$
Grazing deflection $\hat{\alpha}_{\rm Yuk,conf}(b_{\rm graz})$ (rad)	`1.29085207`	`1.33402365`	Conformal null geodesic at each branch's own grazing ray
Fixed-b deflection $\hat{\alpha}_{\rm Yuk,conf}(b =$ 15.064914 km) (rad)	`1.22474419` ($r_0 = 11.917144$ km)	`1.33402365` ($r_0 = 11.736304$ km)	Evaluated at shared exterior $b = b^{\rm (II\text{-}Conf)}_{\rm graz}$ (+8.92% from larger $M_{\rm ADM}$)
Grazing relative shift vs. $\hat{\alpha}_{\rm Schw}(M_{\rm ADM})$ / $\hat{\alpha}_{\rm Schw}(M_{\rm surf})$	`-2.867e-4` / `+3.652e-3`	`-1.671e-4` / `+2.184e-3`	Ray dives inside exterior scalar cloud $(m(r_0) < M_{\rm ADM})$
Far-field relative shift vs. $\hat{\alpha}_{\rm Schw}(M_{\rm ADM})$ ($b \geq 200$ km)	`<= 8.03e-13` (`0.0` at $b \geq R_\odot$)	`<= 1.79e-13` (`0.0` at $b \geq R_\odot$)	Exponential screening $\exp(-b/L) \to 0$ on both branches
Interior–exterior surface slope mismatch $\lvert\phi'_{L,\rm int}(R_*) - \phi'_{L,\rm ext}(R_*)\rvert$ (km^{-1})	`1.88e-09` (4-step Robin)	Robin: `1.465e-09` (4-step) / `3.937e-15` (every-step); FD2 ($N = 4000$): `4.723e-10`	Enforced Robin BC vs. independent one-sided FD2 ($\mathcal{O}(h^2)$)
Converged interior scalar ODE residual $\max_{r\in(0,R_*)}\lvert{\rm res}_{\rm ODE}\rvert$ (km^{-2})	`1.42e-10` (iter) / `1.95e-14` (banded)	`1.078e-10` (iter, `tol=2e-8`) / `6.958e-14` (`tol=1e-11`) / `1.480e-14` (banded)	Discrete residual of 2nd-order banded Helmholtz–Yukawa system

2B.1. Targeted Numerical Verification of the `Branch II-Conf` SLy $\log_{10}(\rho_{0c}) = 15.0$ BVP: Interior Residual, Surface $\phi_L'(R_*)$ Matching, and $N = 2000$ vs. $N = 4000$ Mesh Refinement The script `gw170817_and_lensing_audit.py` (function `compute_branch_ii_conf_bvp_convergence_and_mesh_refinement`, stored under `gw170817_and_lensing_summary.json` → `bvp_convergence_and_mesh_refinement`) directly logs three numerical verification diagnostics for the single examined $L = 10$ km, $\alpha_Y = 1$, SLy $\log_{10}(\rho_{0c}) = 15.0$ `Branch II-Conf` trace-sourced reference star BVP:

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1. **Converged Interior Discrete Equation Residual ($\max_{r\in(0,R_*)}|\mathbf{res_{ODE}}|$, $\mathbf{km^{-2}}$) and Exterior Normalized Collocation Residual:**
   - On the converged $N = 601$ grid, the discrete residual of the interior scalar ODE $B\phi_L'' + (2B/r + B\nu' + B'/2)\phi_L' - \phi_L/L^2 + \alpha_Y^2 e^{-4\alpha_Y\phi_L}(\tilde{\varepsilon} - 3\tilde{P}) = 0$ is $\max|\mathbf{res_{ODE,banded}}| = 1.480 \times 10^{-14}$ $\mathbf{km^{-2}}$ for the direct tridiagonal banded solve (`_solve_curved_scalar_with_k`, which measures the accuracy of solving its own discrete linear system on that grid), $\max|\mathbf{res_{ODE,iter}}| = 1.078 \times 10^{-10}$ $\mathbf{km^{-2}}$ for the relaxed fixed-point iterate (`tol_iter = 2e-8`, 19 iterations, $\max|\phi_L - \phi_{L,\mathrm{banded}}| = 3.065 \times 10^{-9}$), and $\max|\mathbf{res_{ODE,iter}}| = 6.958 \times 10^{-14}$ $\mathbf{km^{-2}}$ on the separate `tol_iter = 1e-11` run with every-step 4-component exterior Robin updates ($N = 601$, 36 iterations). The exterior 4-component BVP (`scipy.integrate.solve_bvp`, `tol = 1e-12`) has maximum normalized RMS collocation residual $\max(\mathbf{rms_residuals}) = 9.998 \times 10^{-13}$ on $[R_*, R_{\max}]$ (a dimensionless normalized collocation residual on the exterior mesh, distinct from the interior km^{-2} discrete ODE residual).

2. **Distinction Between the Enforced Robin Derivative ($\phi'_{L,\mathbf{int,Robin}}(R_*)$) and the Independent One-Sided Finite-Difference Derivative ($\phi'_{L,\mathbf{int,FD2}}(R_*)$) at $r = R_*$:**
   - **Enforced Robin boundary derivative ($\phi'_{L,\mathbf{int,Robin}}(R_*) = -k_{\mathbf{int}}\phi_L(R_*)$):** Under the 4-step Robin update schedule (`it in (1, 3, 6, 10)`, `tol_iter = 2e-8`, $N = 601$), the interior Robin coefficient is $k_{\mathrm{int}} = 0.232931108388$ km^{-1} vs. the exterior 4-component curved Dirichlet-to-Neumann coefficient $k_{\mathrm{curved}} = 0.232931461902$ km^{-1} ($|\Delta k| = 3.535 \times 10^{-7}$ km^{-1}, relative difference 1.518×10^{-6}); the enforced Robin derivative $\phi'_{L,\mathrm{int,Robin}}(R_*) = -0.000965074633$ km^{-1} and exterior $\phi'_{L,\mathrm{ext,curved}}(R_*) = -0.000965076098$ km^{-1} differ by $|\Delta\phi'_{L,\mathbf{Robin}}(R_*)| = 1.465 \times 10^{-9}$ $\mathbf{km^{-1}}$. When the curved Robin coefficient is updated in **every outer iteration** (`it >= 1`) from the 4-component exterior BVP (`tol_iter = 1e-11`, $N = 601$, 36 iterations), $k_{\mathrm{int}} = 0.232931462480$ km^{-1} and $k_{\mathrm{curved}} = 0.232931462481$ km^{-1} ($|\Delta k| = 9.502 \times 10^{-13}$ $\mathbf{km^{-1}}$), so the **enforced Robin derivative** matches to $|\phi'_{L,\mathbf{int,Robin}}(R_*) - \phi'_{L,\mathbf{ext,curved}}(R_*)| = 3.937 \times 10^{-15}$ $\mathbf{km^{-1}}$ (and $|\phi'_{L,\mathrm{int,ghost}}(R_*) - \phi'_{L,\mathrm{ext,curved}}(R_*)| = 1.430 \times 10^{-14}$ km^{-1} for the ghost-point centered interior derivative), while shifting R_* and M_{ADM} by only $|\Delta R_*| = 2.06 \times 10^{-7}$ km and $|\Delta M_{\mathrm{ADM}}| = 6.07 \times 10^{-8}\ M_\odot$.
   - **Independent 3-point one-sided second-order finite-difference derivative ($\phi'_{L,\mathbf{int,FD2}}(R_*) = (3\phi_{N-1} - 4\phi_{N-2} + \phi_{N-3})/(2h)$):** Separated from the enforced Robin condition, the one-sided FD2 derivative differs from $\phi'_{L,\mathrm{ext,curved}}(R_*)$ by 2.1213×10^{-8} km^{-1} at $N = 601$, 1.9048×10^{-9} km^{-1} at $N = 2000$, and 4.7228×10^{-10} $\mathbf{km^{-1}}$ **at** $N = 4000$ ($\approx 4.89 \times 10^{-7}$ relative to the slope $|\phi'_L(R_*)| = 9.6507 \times 10^{-4}$ km^{-1}), and converges in second order ($\mathcal{O}(h^2)$) as the mesh is refined ($N = 2000 \to 4000$ empirical improvement ratio: `4.0332`, consistent with the expected second-order mesh scaling $(h_{2000}/h_{4000})^2 = 4.0020$).

3. **Mesh Refinement Test ($N = 601$ vs. $N = 2000$ vs. $N = 4000$, `run_variant: DOP853`, `rtol_ivp = 1e-11`, `atol = 1e-13`, `max_step = 0.02 km`, `tol_iter = 1e-10`, Every-Step Robin Update):**

Grid N	$h = \Delta r$ (km)	R_* (km)	M_{surf} ($M_\odot$)	M_{ADM} ($M_\odot$)	$\phi_L(0)$	$\phi_L(R_*)$	$\phi'_{L,\mathrm{int,Robin}}(R_*)$ (km^{-1})	$\phi'_{L,\mathrm{ext,curved}}(R_*)$ (km^{-1})	$\lvert\phi'_{L,\mathrm{int,FD2}} - \phi'_{L,\mathrm{ext}}\rvert(R_*)$ (km^{-1})	$\max\lvert\mathrm{res}_{\mathrm{ODE,iter}}\rvert$ (km^{-2})	$\hat{\alpha}_{\mathrm{tide,conf}}(b_{\mathrm{graz}})$ (rad)
601	0.019438	11.662698	2.156063544	4.562034848	0.011576807	0.004143168	−0.000965074633	−0.000965076098	7.422109652e-08	8.568e-13	1.334023631
2000	0.005834	11.662697	4.956063527	7.562034651	0.011576803	0.004143134	−0.000965076098	−0.000965076098	1.686343e-09	9.844e-13	1.334023603
4000	0.002916	11.662697	4.456063526	6.562034640	0.011576803	0.004143132	−0.000965076098	−0.000965076098	4.652874e-10	9.430e-13	1.334023601
$\lvert\Delta\rvert_{2000\to4000}$	—	4.213e-08 (3.61e-9)	9.670e-09 (6.20e-9)	1.097e-08 (7.03e-9)	2.042e-09 (1.76e-7)	1.920e-09 (4.63e-7)	4.463e-10	4.463e-10	**Scaling:** 4.0332 (expected $\mathcal{O}(h^2)$: 4.0020)	—	1.055e-09 (7.91e-10)

Table 3: Dedicated 1.000000 $M_\odot$ **Solar–Cassini Benchmark ($GM_\odot/c^2 = 1.4766250385$ km, $R_\odot = 695700.0$ km, $r_E = 1$ AU, $r_S = 8.43$ AU; `source_body = Sun_1p00Msun`)**

Row (`scale_label`)	b (km)	$b/R_\odot$	$\hat{\alpha}_{\mathrm{exact}}$ ($''$)	$\hat{\alpha}_{\mathrm{1PN}}$ ($''$)	$\Delta t_{\mathrm{Shapiro}}(10\,r_\oplus)$ (μs)	$\Delta t_{\mathrm{Cassini,rt}}$ (μs)	$\Delta t_{\mathrm{Cassini,rt}}$ with geometric -1 (μs)	$\log_{10}\lvert\gamma_{\mathrm{PPN}} - 1\rvert$
`Solar_1Msun_1p05Rsun`	1.463010e+05	1.0500	1.751201	1.751190	67.8831	280.9434	261.2415	-30214.67
`Solar_1Msun_Cassini_1p06Rsun`	1.477120e+06	1.0600	1.094498	1.094494	67.8830	262.4234	242.7215	-48342.99
`Solar_1Msun_0p25Rsun`	1.739250e+05	0.2500	0.175119	0.175119	67.8829	—	—	-302139.47

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Row (scale_label)	b (km)	$b/R_\odot$	$\hat{\alpha}_{\rm exact}$ ($''$)	$\hat{\alpha}_{\rm 1PN}$ ($''$)	$\Delta t_{\rm Shapiro}(10\,r_s)$ (μs)	$\Delta t_{\rm Cassini,rt}$ (μs)	$\Delta t_{\rm Cassini,rt}$ with geometric -1 (μs)	$\log_{10}\lvert\gamma_{\rm PPN}-1\rvert$
Solar_1Msun_1AU	1.495979e+08	215.0322	0.008144	0.008144	67.8829	—	—	-6496953.77

1.3.4 3.4. Weak-Lensing Convergence ($\kappa(\theta)$) on Galactic and Cluster Scales: Bridge to Module 2 (Bullet Cluster)

The weak-lensing convergence ($\kappa(\theta) = \frac{1}{2}\nabla_\theta^2\psi(\theta)$) is governed by the line-of-sight integral of the Weyl potential ($\Phi_{\rm lens} = \frac{1}{2}(\tilde{\Psi}+\tilde{\Phi})$):

$$\kappa(\mathbf{b}_\perp) \;=\; \frac{1}{2\,c^2}\,\frac{D_L D_{LS}}{D_S}\int_{-\infty}^{+\infty}\nabla_\perp^2\big(\tilde{\Psi}(\mathbf{b}_\perp,z)+\tilde{\Phi}(\mathbf{b}_\perp,z)\big)\,dz \;=\; \frac{\Sigma_{\rm lens}(\mathbf{b}_\perp)}{\Sigma_{\rm crit}}, \qquad \Sigma_{\rm crit} \;=\; \frac{c^2}{4\pi G}\,\frac{D_S}{D_L D_{LS}}.$$

Let us evaluate what the fixed parameter pair $L = 10$ km, $\alpha_Y = 1$ predicts on galactic (~ 30 kpc, SPARC NGC 3198) and galaxy-cluster (~ 250 kpc, Bullet Cluster 1E 0657-558) scales (`gw170817_and_lensing_summary.json`):

1. **Behavior of the Local Fixed-$L = 10$ km Yukawa Component on kpc–Mpc Scales:**
   - **Exterior field of compact sources (stars, remnants):** At $b = 1$ kpc ($b/L = 3.086\times10^{15}$) and $b = 250$ kpc ($b/L = 7.714\times10^{17}$), the exponential screening factor $e^{-b/L} = 10^{-3.35\times10^{17}} = 0$ exponentially suppresses to zero any local $L = 10$ km Yukawa tail.
   - **Equilibrium local field inside diffuse baryonic media (galactic disk or X-ray intracluster plasma, $\rho_{\rm ICM} \sim 10^{-22}$ kg/m^3):** In a quasi-homogeneous background of density ρ, the local Helmholtz–Yukawa equilibrium field is $\phi_L^{\rm local}(L = 10\text{ km}) = \alpha_Y\gamma_L\rho$:
     - In the outer disk of NGC 3198 ($\rho_{\rm gal} \sim 10^{-24}$ kg/m^3, $R = 30.08$ kpc):

       $$\phi_L^{\rm local}(10\text{ km}) \;=\; 7.426\times10^{-44}, \qquad \frac{|\Phi_N|}{c^2} \;\approx\; \frac{V_{\rm bar}^2}{c^2} \;=\; 5.499\times10^{-8} \quad\Longrightarrow\quad \frac{c^2\phi_L^{\rm local}(10\text{ km})}{|\Phi_N|} \;=\; 1.350\times10^{-36}.$$

     - In the hot intracluster medium of the Bullet Cluster ($\rho_{\rm ICM} \sim 10^{-22}$ kg/m^3, $b = 250$ kpc):

       $$\phi_L^{\rm local}(10\text{ km}) \;=\; 7.426\times10^{-42} \quad\Longrightarrow\quad \kappa_{\rm lens}^{\rm local\ L=10\ km}(\mathbf{b}_\perp) \;=\; \kappa_{\rm baryon}(\mathbf{b}_\perp) \quad \left(\frac{\kappa_{\rm lens}^{\rm local\ L=10\ km}}{\kappa_{\rm baryon}} = 1.000000000000\right)$$

2. **Physical Conclusion and Boundary Condition for Module 2 (Bullet Cluster 1E 0657-558):**
   - The local $L = 10$ km, $\alpha_Y = 1$ Yukawa interaction is strictly a **strong-field / compact-object ($r \lesssim 100$ km) fifth force** that:
     - satisfies the GW170817 $c_T = c$ bound identically ($(c_T - c)/c = 0$),
     - suppresses scalar dipolar and monopolar radiation across binary-pulsar and LIGO inspiral frequencies via the 4771.35 Hz mass cutoff,
     - satisfies the Solar-System Cassini bound at $b = 1.6\,R_\odot$ with $|1-\gamma_{\rm PPN}| = 0 < 2.3\times10^{-5}$,
     - and simultaneously produces **zero non-GR contribution** on kiloparsec and megaparsec scales ($b/L \sim 10^{16}$–10^{18}), neither to galactic rotation curves (where NGC 3198 at $R = 30.08$ kpc requires a dynamical boost $C_{\rm req} = (V_{\rm obs}/V_{\rm bar})^2 = 4.311$ at $\Upsilon_* = 0.5$; `step6`) nor to the Bullet Cluster weak-lensing convergence map (where the galaxy subcluster peak exhibits $\kappa_{\rm obs}/\kappa_{\rm baryon,gal} \approx 0.232/0.0258 \approx 9.0$ at $\Sigma_{\rm crit} \approx 3.10\times10^9\ M_\odot\,{\rm kpc}^{-2}$).
   - For a macroscopic clock field ($\phi_{\rm macro}$, such as the chameleon/nonlocal $L_{\rm eff}(\rho) \sim 10$–$300$ kpc branch analyzed in `step6`, `step11`, `step14`) to enhance weak-lensing convergence ($\kappa > \kappa_{\rm baryon}$), **two conditions must hold simultaneously**:
     1. The metric coupling cannot be purely conformal ($\tilde{g}_{\mu\nu} = A^2(\phi)g_{\mu\nu}$), because the conformal factor cancels in the Weyl potential $(\tilde{\Psi}+\tilde{\Phi})/2$; it requires a **disformal / refractive coupling** ($\tilde{g}_{\mu\nu} = A^2(\phi)g_{\mu\nu} + B(\phi)\partial_\mu\phi\partial_\nu\phi$ or effective optical metric potential $\Phi_{\rm lens}^{\rm eff} = \Phi_N - \frac{1}{2}c^2\phi_{\rm macro}$, `step14`); and
     2. The effective range of the scalar field must be environment-dependent ($L_{\rm eff}(\rho) \propto \rho^{-1/2}$) or nonlocal, since the fixed $L = 10$ km Yukawa kernel vanishes identically for $b \gg 10$ km.

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2 Step 24 — Bullet Cluster (1E 0657-558) Gravitational Lensing Convergence $\kappa(\theta)$ and Gas vs. Galaxy Peak Offset Analysis at $L = 10$ km, $\alpha_Y = 1$ (step24_en, V26)

Source file: step24_bullet_cluster_en.md

2.1 I. Physical and Methodological Framework: Synthetic 1D Illustrative Collision-Axis Test of the Bullet Cluster (1E 0657-558)

In step23_gw170817_and_lensing_en.md, we proved that within the canonical V24 (10.5281/zenodo.23680715) and V25 (10.5281/zenodo.23699685) short-range scalar–tensor Yukawa baseline ($L = 10$ km $= 10^4$ m, $\alpha_Y = 1$, $\beta_{\rm Yuk} = 2\tilde{\alpha}_L\alpha_Y^2 = \alpha_Y^3/(4\pi) = 1/(4\pi) \approx 0.0795775$), the tensor gravitational-wave propagation speed on the purely conformal minimal branch is identically $c_T = c$ ($\Delta c_T/c = 0$), and under the conformal Jordan metric ($\tilde{g}_{\mu\nu} = A^2(\phi_L)g_{\mu\nu} = e^{-2\alpha_Y\phi_L}g_{\mu\nu}$) the linear Weyl lensing potential satisfies $\Phi_{\rm lens} = \frac{1}{2}(\tilde{\Psi} + \tilde{\Phi}) = \Phi_N$. In Step 24 (step24), we apply this framework to a **synthetic 1D illustrative collision-axis model** of the **Bullet Cluster (1E 0657-558)** (Markevitch et al. 2004, *ApJ* 606, 819; Clowe et al. 2006, *ApJ* 648, L109; Bradač et al. 2006, *ApJ* 652, 937).

2.1.1 1.1. Cosmological and Lensing Parameters (1E 0657-558)

- **Lens redshift:** $z_L = 0.296$, angular diameter distance $D_L = 910.0$ Mpc ($1' \leftrightarrow 264.7$ kpc).
- **Characteristic background source redshift:** $z_S \approx 0.8$–1.0 (fiducial value $z_S = 0.85$).
- **Critical surface mass density:**

$$\Sigma_{\rm crit} = \frac{c^2 D_S}{4\pi G D_L D_{LS}} = 3.10 \times 10^9\ M_\odot\,{\rm kpc}^{-2} = 0.6474\ {\rm g\,cm}^{-2}.$$

- **Supersonic merger shock velocity:** $v_{\rm coll} \approx 4500$ km/s $\approx 0.01501\,c$, driving ram-pressure stripping that spatially separates the collisional X-ray intracluster medium (ICM) from the collisionless galaxies.

2.1.2 1.2. Four-Component Baryonic Model Along the Collision Axis ($x \in [-800, +800]$ kpc)

Motivated by the characteristic mass and scale ratios from Chandra X-ray surface brightness maps and HST/VLT/Magellan optical imaging, the baryonic mass distribution along the collision axis ($y = 0$, $x \in [-800, +800]$ kpc) is represented by four analytic components with aperture-normalized ($R_{\rm ap} = 650.0$ kpc) King / modified-Hubble ($\beta = 1$) surface density profiles $\Sigma_i(R) = \Sigma_{0,i}(1 + R^2/R_{c,i}^2)^{-1}$, where $\Sigma_{0,i} = \frac{M_i}{\pi R_{c,i}^2 \ln(1+R_{\rm ap}^2/R_{c,i}^2)}$:

Component ID	Physical Description	Centroid $x_{c,i}$ (kpc)	Aperture Mass M_i ($M_\odot$)	Core Radius $R_{c,i}$ (kpc)	Central $\Sigma_{0,i}$ ($M_\odot$/kpc^2)	Central $\kappa_{0,i} = \Sigma_{0,i}/\Sigma_{\rm crit}$
gal_main	Main Cluster Collisionless Galaxies (Stars + Compact Remnants)	-350.0	5.50e12	120.0	3.9329e7	0.012687
gas_main	Main Cluster Collisional X-ray ICM Gas (Chandra, ram-pressure stripped)	-170.0	2.60e13	150.0	1.3638e8	0.043995

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Component ID	Physical Description	Centroid $x_{c,i}$ (kpc)	Aperture Mass M_i ($M_\odot$)	Core Radius $R_{c,i}$ (kpc)	Central $\Sigma_{0,i}$ ($M_\odot$/kpc^2)	Central $\kappa_{0,i} = \Sigma_{0,i}/\Sigma_{\rm crit}$
`gas_sub`	Subcluster ("Bullet") Shocked X-ray ICM Gas	`+140.0`	`6.50e12`	`75.0`	`8.4628e7`	`0.027300`
`gal_sub`	Subcluster ("Bullet") Collisionless Galaxies (160 kpc ahead of shock)	`+300.0`	`1.80e12`	`65.0`	`2.9296e7`	`0.009450`

- **Total baryonic mass budget in the aperture:**

$$M_{\rm gas,tot} = 3.25\times10^{13}\ M_\odot, \qquad M_{\rm gal,tot} = 7.30\times10^{12}\ M_\odot, \qquad M_{\rm bary,tot} = 3.98\times10^{13}\ M_\odot, \qquad \frac{M_{\rm gas,tot}}{M_{\rm gal,tot}} = 4.4521.$$

- **Observationally motivated peak positions and synthetic 1D reference curve (`kappa_synth_target_1D`):** In weak + strong lensing reconstructions (Clowe et al. 2006; Bradač et al. 2006), the convergence peaks lie near the collisionless galaxy centroids ($x^{\rm obs}_{\kappa,\rm main} \approx -350.0$ kpc, $\kappa^{\rm obs}_{\rm peak,main} \approx 0.320$; $x^{\rm obs}_{\kappa,\rm sub} \approx +300.0$ kpc, $\kappa^{\rm obs}_{\rm peak,sub} \approx 0.220$) rather than the dominant X-ray gas centroids ($x = -170$ kpc and $+140$ kpc). **Methodological Scope and Caveat (V26):** The `kappa_synth_target_1D` column in `bullet_cluster_convergence_profile.csv` is **not an observational data series**, but a **self-constructed 1D synthetic reference curve**:

$$\kappa_{\rm synth}(x) = \kappa_{\rm GR,bary}(x) + \frac{w_{\rm synth,main}\,\Sigma_{\rm gal,main}(x) + w_{\rm synth,sub}\,\Sigma_{\rm gal,sub}(x)}{\Sigma_{\rm crit}}, \qquad (w_{\rm synth,main} = 22.9765,\ w_{\rm synth,sub} = 21.3095),$$

built from the model's own four King profiles with two galaxy weights tuned to reproduce the illustrative peak heights at $x = -350$ kpc ($\kappa = 0.320$) and $x = +300$ kpc ($\kappa = 0.220$). Accordingly, the metric reported in `synth_mse_norm_vs_target`,

$$M_{\rm synth} \equiv \frac{1}{N}\sum_{i=1}^{N}\left[\frac{\kappa(x_i) - \kappa_{\rm synth}(x_i)}{\sigma_{\rm ref}}\right]^2 \qquad (\sigma_{\rm ref} = 0.020,\ N = 321),$$

is a **normalized mean-squared deviation from this synthetic 1D target curve**, **NOT** an observational χ^2_ν fit to Clowe et al. (2006) or Bradač et al. (2006) data. A genuine observational χ^2 fit requires a published 2D convergence map, full error covariance matrix, pre-registered spatial selection, and proper subtraction of fitted degrees of freedom ($N_{\rm dof} = N_{\rm bins} - p$).

2.2 II. Test 1 — Pure Local Single-Scale Yukawa Model at Fixed $L = 10$ km, $\alpha_Y = 1$ (Strict Cluster-Scale Falsification / No-Go Result)

We first evaluate the exact 3D scalar field $\phi_L^{\rm local}(\mathbf{r})$ and projected 2D lensing convergence $\kappa_{\rm local}(x)$ produced by the V24/V25 local Yukawa model at fixed $L = 10$ km $= 3.24077929 \times 10^{-16}$ kpc and $\alpha_Y = 1$.

2.2.1 2.1. Algebraic Contact Limit on Cluster Scales ($R_c \sim 65$–150 kpc $\sim (2.0$–$4.6)\times10^{17}\,L$)

In the V23–V25 sign convention ($\phi_L > 0$ near matter sources), the static 3D Helmholtz–Yukawa field equation is:

$$\left(\nabla^2 - \frac{1}{L^2}\right)\phi_L(\mathbf{r}) = -\alpha_{\rm std}\,\frac{4\pi G}{c^2}\,\rho_b(\mathbf{r}), \qquad \alpha_{\rm std} = \frac{\alpha_Y}{4\pi} \approx 0.0795775.$$

Because the cluster core radii satisfy $R_{c,i} \in [65, 150]$ kpc, the dimensionless scale ratio is:

$$\frac{R_{c,i}}{L} \in \left[2.006\times10^{17},\ 4.628\times10^{17}\right] \implies \left(\frac{L}{R_{c,i}}\right)^2 \in \left[4.668\times10^{-36},\ 2.486\times10^{-35}\right].$$

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Spatial derivatives ($|\nabla^2\phi_L| \sim R_c^{-2}|\phi_L|$) are suppressed by more than 35 orders of magnitude relative to the mass term $-L^{-2}\phi_L$, reducing the 3D scalar field to the algebraic contact limit:

$$\left|\phi_L^{\rm local}(\mathbf{r})\right| \simeq \alpha_{\rm std}\, L^2\, \frac{4\pi G}{c^2}\, \rho_b(\mathbf{r}) \in \left[10^{-44},\ 3.086\times 10^{-42}\right].$$

For individual compact objects and stars inside the galaxies, each star's $L = 10$ km Yukawa halo decays as $e^{-r/(10\ \rm km)}$, falling below $e^{-10} \approx 4.5\times 10^{-5}$ within 100 km ($< 1.5\times 10^{-4}\, R_\odot$, or $< 3\times 10^{-18}$ kpc) and vanishing as $\exp(-10^{17}) = 0$ on kiloparsec scales.

2.2.2 2.2. Conformal vs. Disformal Lensing Convergence at $L = 10$ km

1. **Conformal Jordan Branch (`step14` Branch II-Conf, `step23`):** Under $\tilde g_{\mu\nu} = A^2(\phi_L) g_{\mu\nu} = e^{-2\alpha_Y \phi_L} g_{\mu\nu}$, the scalar field cancels identically out of the linear Weyl potential ($\Phi_{\rm lens}^{\rm conf} = \Phi_N$), while the exterior scalar energy cloud of compact objects ($\Delta M_{\phi,\rm ext}^{\rm curved}/M_{\rm surf} = +1.55\times 10^{-3}$ for neutron stars, and $\lesssim 10^{-16}$ for main-sequence stars) is already absorbed into the stars' asymptotic ADM masses $M_{\rm gal}$. Consequently:

$$\Delta\kappa^{\rm conf}_{\rm local\ \ L=10\ km}(x) = 0 \quad \text{(identically)}.$$

2. **Disformal Refractive Clock Branch (`step14` Branch II-Disf):** If on a separate disformal optical branch $\Phi_{\rm lens}^{\rm disf} = \Phi_N - \frac{1}{2}c^2\phi_L$, integrating along the line of sight $z \in (-\infty, +\infty)$ and applying the 2D transverse Laplacian yields the 2D Fourier filter $\frac{k_\perp^2}{k_\perp^2 + L^{-2}} \simeq L^2 k_\perp^2 \leftrightarrow -L^2\nabla_\perp^2$, giving:

$$\Delta\kappa^{\rm disf}_{\rm local\ \ L=10\ km}(x) = -\frac{\alpha_{\rm std}}{2}\, L^2\, \nabla_\perp^2 \kappa_{\rm GR,bary}(x), \qquad \max_x \left|\frac{\Delta\kappa^{\rm disf}_{\rm local}}{\kappa_{\rm GR,bary}}\right| = 2.171\times 10^{-36} \quad (\text{and } < 10^{-40} \text{ in the outer halo}).$$

2.2.3 2.3. Transparent Scientific Verdict for Test 1

On both metric coupling branches:

$$\kappa_{\rm local\ \ L=10\ km}(x) = \kappa_{\rm GR,bary}(x) = \frac{\Sigma_{\rm gas}(x) + \Sigma_{\rm gal}(x)}{\Sigma_{\rm crit}}.$$

- The convergence peaks remain locked to the **X-ray gas centroids**: in the 4-component global superposition they sit at $x^{\rm local}_{\rm peak,main} = -175.44$ kpc ($\kappa^{\rm local}_{\rm peak,main} = 0.04503$) and $x^{\rm local}_{\rm peak,sub} = +137.01$ kpc ($\kappa^{\rm local}_{\rm peak,sub} = 0.03697$), or in the shock-separated / steeper-outer-gas subcluster-pair decomposition at $x^{\rm local}_{\rm peak,main} = -170.7$ kpc and $x^{\rm local}_{\rm peak,sub} = +142.3$ kpc ($\kappa^{\rm local}_{\rm peak,main} = 0.0450$).
- The main peak amplitude is 7.11× **smaller** than the $\kappa = 0.320$ reference peak, and the peaks are **displaced by** 157.7–179.3 **kpc** from the galaxy centroids (−350 kpc and +300 kpc) toward the collisional X-ray gas ($M_{\rm synth} = 32.089$).
- **Conclusion: At fixed $L = 10$ km and $\alpha_Y = 1$, the short-range Yukawa fifth force is purely a compact-object/neutron-star effect and produces zero macroscopic modification to Bullet Cluster lensing ($\Delta\kappa/\kappa_{\rm bary} \le 2.17\times 10^{-36}$; $< 10^{-40}$ in the outer halo).**

2.3 III. Test 2 — Phenomenological Macroscopic Extensions on the Synthetic 1D Test: What It Takes to Move the $\kappa(x)$ Peak from the X-ray Gas to the Galaxies

We now examine quantitatively within the synthetic 1D King model (`step11`, `step14` Branch II-Disf: $\Phi_{\rm lens} = \Phi_N - \frac{1}{2}c^2\phi_{\rm macro}$, with illustrative cluster Helmholtz scale $L_{\rm macro} = 300.0$ kpc) the two macroscopic sourcing cases.

2.3.1 3.1. Case 2A (`Macro_unscreened_bulk_baryon`) — Static Macroscopic Kernel Depending Only on Smoothed Bulk Baryon Density ($\bar\rho_b = \rho_{\rm gas} + \rho_{\rm gal}$)

Suppose a macroscopic scalar field is sourced universally by total baryon density with a uniform coupling $\eta_{\rm bulk}$:

$$\left(\nabla^2 - L_{\rm macro}^{-2}\right)\phi_{\rm macro}(\mathbf{r}) = -\frac{4\pi G}{c^2}\,\eta_{\rm bulk}\left[\rho_{\rm gas}(\mathbf{r}) + \rho_{\rm gal}(\mathbf{r})\right], \qquad \widetilde\Sigma_{\phi,\rm bulk}(\mathbf{k}_\perp) = \eta_{\rm bulk}\,\frac{k_\perp^2}{k_\perp^2 + L_{\rm macro}^{-2}}\,\widetilde\Sigma_b(\mathbf{k}_\perp).$$

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Calibrating $\eta_{\rm bulk} = 8.6523$ boosts the main convergence peak by 7.11× to match $\kappa_{\rm peak,main} = 0.3200$. **However**, because $M_{\rm gas,tot}/M_{\rm gal,tot} = 4.4521$, the convergence peaks **remain locked to the X-ray gas**:

$x^{\rm bulk}_{\rm peak,main} = -176.02$ kpc (offset from galaxies: +173.98 kpc), $x^{\rm bulk}_{\rm peak,sub} = +138.35$ kpc (offset from galaxies: −161.65 kpc

($M_{\rm synth} = 20.723$). This demonstrates that **no static modified-gravity or scalar–tensor kernel sourced solely by smoothed bulk baryon density ($\rho_{\rm gas} + \rho_{\rm gal}$) can shift the convergence peaks from the dominant X-ray gas to the collisionless galaxies.**

2.3.2 3.2. Case 2B (`Macro_collisionless_compact_clock`) — Phenomenological Collisionless / Compact-Well Sourcing Hypothesis

For a macroscopic field to center its lensing convergence peaks on the collisionless galaxies without cold dark matter (CDM) particles, the macroscopic sector would need one of two physical properties:

1. **Hypothesis 2B-1 — Compactness / Deep-Potential-Well Clock Sourcing:** In the diffuse X-ray ICM plasma ($n_e \sim 10^{-3}$ cm^{-3}), small-scale metric-gradient variance vanishes ($\langle(\nabla\nu)^2\rangle_{\rm micro} \approx 0$), whereas in the galaxy component baryons are bound inside stellar and compact-remnant potential wells. If the effective source of a macroscopic clock field is weighted by small-scale gradient energy or local compactness, it couples predominantly to the collisionless ($\sigma_{\rm coll} \approx 0$) stellar/compact-remnant population.
2. **Hypothesis 2B-2 — Non-Equilibrium / Retarded Scalar Memory:** If the macroscopic field has a dynamical relaxation timescale $\tau_{\rm relax} \gtrsim 150$ Myr longer than the core-passage time $t_{\rm cross} = \Delta x/v_{\rm coll} \approx 180$ kpc/(4500 km/s) $\approx$ 39.1 Myr, the scalar halo can travel ballistically with the collisionless galaxies during the supersonic merger.

2.3.3 3.3. Model-Dependent Threshold Calculation for $\eta_{\rm gal}/\eta_{\rm gas}$ and Methodological Caveats

With phenomenological macroscopic refractive coupling weights $\eta_{\rm gal}$ and $\eta_{\rm gas}$ in the chosen 1D King geometry ($R_{c,i} \in [65, 150]$ kpc) at $L_{\rm macro} = 300.0$ kpc:

1. **Peak-Dominance Threshold ($\kappa(-350\ \mathbf{kpc}) \geq \kappa(-170\ \mathbf{kpc})$ and $\kappa(+300\ \mathbf{kpc}) \geq \kappa(+140\ \mathbf{kpc})$):**
   - Direct surface-density ratio (without Helmholtz filtering):
   $$\frac{\Sigma_{\rm gas,main}(-170)}{\Sigma_{\rm gal,main}(-350)} = 3.4592, \qquad \frac{\Sigma_{\rm gas,sub}(+140)}{\Sigma_{\rm gal,sub}(+300)} = 2.8895.$$
   - Including $L_{\rm macro} = 300$ kpc Helmholtz–Yukawa filtering (at reference $\eta_{\rm gas} = 1$):
   $$\left(\frac{\eta_{\rm gal}}{\eta_{\rm gas}}\right)_{\rm dom,main} = 5.1204, \qquad \left(\frac{\eta_{\rm gal}}{\eta_{\rm gas}}\right)_{\rm dom,sub} = 5.0972.$$
2. **< 5.0 kpc Centroid-Offset Threshold ($|x_{\mathbf{peak,main}} - (-350\ \mathbf{kpc})| \leq 5.0\ \mathbf{kpc}$):**
   - To prevent the X-ray gas tail in this 1D King profile from pulling the main convergence peak by more than 5.0 kpc away from the galaxy centroid at −350.0 kpc, the required ratio is:
   $$\left(\frac{\eta_{\rm gal}}{\eta_{\rm gas}}\right)_{|\Delta x| \leq 5\ \rm kpc} \geq 25.6650.$$
3. **Calibrated Case 2B Illustrative Model (`Macro_collisionless_compact_clock`, $\eta_{\rm gas} = 0$, $\eta_{\rm gal,main} = 31.7448$, $\eta_{\rm gal,sub} = 24.1556$):**
   - Main cluster peak: $x_{\rm peak,main} = -347.86$ kpc (centroid offset from galaxies: $+2.14\ \mathbf{kpc} < 5\ \mathbf{kpc}$), $\kappa_{\rm peak,main} = 0.32011$ ($\kappa(-350\ {\rm kpc}) = 0.32000$).
   - Subcluster peak: $x_{\rm peak,sub} = +299.45$ kpc (centroid offset from galaxies: $-0.55\ \mathbf{kpc} < 5\ \mathbf{kpc}$), $\kappa_{\rm peak,sub} = 0.22002$ ($\kappa(+300\ {\rm kpc}) = 0.22000$).
   - Deviation from the synthetic 1D target curve (`kappa_synth_target_1D`): RMS($\Delta\kappa$) = 0.02559, normalized synthetic mean-squared deviation $M_{\rm synth} = 1.637$ (compared with $M_{\rm synth} = 32.089$ for the local $L = 10$ km model and $M_{\rm synth} = 20.723$ for the Case 2A bulk-baryon model).
4. **Strict Theoretical and Methodological Delineation (V26):**

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- The peak-dominance ratio $\eta_{\rm gal}/\eta_{\rm gas} \geq 5.1204$ and the < 5 kpc centroid-offset ratio $\eta_{\rm gal}/\eta_{\rm gas} \geq 25.6650$ are **model-dependent thresholds specific to the chosen 1D King geometry ($R_{c,i} = 65$–150 kpc) and Helmholtz scale $L_{\rm macro} = 300$ kpc**, not a general model-independent physical theorem.
- Setting $\eta_{\rm gas} = 0$ and fitting two separate galactic weights ($\eta_{\rm gal,main} = 31.7448 \neq \eta_{\rm gal,sub} = 24.1556$) are **phenomenological, illustrative assumptions that are not derived from the covariant scalar–tensor action**.

2.4 IV. Synthetic 1D Illustrative Peak-Offset and Convergence Profile Tables

2.4.1 4.1. Summary of Peak Positions, Amplitudes, and Deviations from the Synthetic 1D Target Curve Across All Four Models (`bullet_cluster_peak_offsets_summary.csv` — Synthetic 1D Illustrative Test)

Model (`model_name`)	$x_{\rm peak,main}$ (kpc)	$\kappa_{\rm peak,main}$	$x_{\rm peak,sub}$ (kpc)	$\kappa_{\rm peak,sub}$	$\Delta x_{\rm main,gal}$ (kpc)	$\Delta x_{\rm main,gas}$ (kpc)	$\Delta x_{\rm sub,gal}$ (kpc)	$\Delta x_{\rm sub,gas}$ (kpc)	RMS $\Delta\kappa_{\rm synth}$	$M_{\rm synth}$ (`synth_mse_norm`
GR_baryon	-175.44	0.04503	+137.01	0.03697	+174.56	-5.44	-162.99	-2.99	0.11329	32.089
TAL_local_Yukawa_DM	-175.44	0.04503	+137.01	0.03697	+174.56	-5.44	-162.99	-2.99	0.11329	32.089
Macro_unscreened_DM	-176.02	0.32002	+138.35	0.26531	+173.98	-6.02	-161.65	-1.65	0.09104	20.723
Macro_collisionless_compact	-147.86	0.32011	+299.45	0.22002	+2.14	-177.86	-0.55	+159.45	0.02559	1.637

(Note: $M_{synth} = \frac{1}{N}\sum_{i=1}^{N}[(\kappa(x_i) - \kappa_{synth}(x_i))/\sigma_{ref}]^2$ is the normalized mean-squared deviation from the self-constructed 1D `kappa_synth_target_1D` reference curve with $\sigma_{ref} = 0.020$, not an observational χ^2_ν.)

2.4.2 4.2. Projected Surface Density and $\kappa(x)$ Profile at Representative Collision-Axis Positions (`bullet_cluster_convergence_profile.csv` — Synthetic 1D Illustrative Test)

x (kpc)	$\Sigma_{\rm gas}$ ($M_\odot$/kpc^2)	$\Sigma_{\rm gal}$ ($M_\odot$/kpc^2)	$\kappa_{\rm GR,bary}$	$\kappa_{\rm local}$ L=10 km	$\Delta\kappa_{\rm local,disf}$	$\kappa_{\rm Macro,2A}$ (bulk)	$\kappa_{\rm Macro,2B}$ (clock)	$\kappa_{\rm synth,1D}$ (`kappa_synth_targ`
-600.0	1.4233e+07	6.8239e+06	0.006793	0.006793	-5.740e-40	0.010277	0.028049	0.062017
-450.0	2.8832e+07	2.1245e+07	0.016154	0.016154	+2.251e-42	0.076874	0.165259	0.188492
-350.0	5.2453e+07	3.5918e+07	0.028507	0.028507	+1.240e-38	0.178665	0.320000	0.320000
-260.0	9.3503e+07	2.3192e+07	0.037644	0.037644	+8.951e-39	0.254340	0.203216	0.225535
-175.0	1.2766e+08	1.1934e+07	0.045031	0.045031	+2.869e-38	0.320000	0.104232	0.141178
-170.0	1.2794e+08	1.1514e+07	0.044984	0.044984	+2.882e-38	0.319487	0.100405	0.137693
-80.0	9.9460e+07	6.7119e+06	0.034249	0.034249	+6.544e-39	0.219456	0.048876	0.087474
+0.0	7.2882e+07	5.0651e+06	0.025144	0.025144	-3.132e-39	0.138610	0.027722	0.064214
+70.0	7.9995e+07	4.8621e+06	0.027373	0.027373	+3.408e-40	0.166192	0.029235	0.063429
+140.0	1.0829e+08	6.1784e+06	0.036924	0.036924	+8.015e-38	0.265185	0.049209	0.080507
+220.0	5.5598e+07	1.3197e+07	0.022192	0.022192	-2.748e-39	0.134507	0.088322	0.110826
+300.0	2.6688e+07	3.0558e+07	0.018466	0.018466	+3.559e-38	0.110815	0.220000	0.220000
+450.0	1.1510e+07	5.4294e+06	0.005464	0.005464	-9.488e-40	0.006652	0.021648	0.042188
+600.0	6.7045e+06	1.8771e+06	0.002768	0.002768	-1.602e-40	-0.004653	0.000158	0.015925

2.5 V. Reproducibility and Generated Artifacts

- **Numerical audit script:** `bullet_cluster_lensing_audit.py` (synced to `code/`).
- **1D collision-axis synthetic convergence profile (321 grid points, $x \in [-800, +800]$ kpc):** `bullet_cluster_convergence_profile.csv` (synced to root, `code/`, `csv/`, and `data/v23/`).
- **Peak-offset and normalized synthetic deviation ($M_{\rm synth}$) summary table:** `bullet_cluster_peak_offsets_summary` (synced to root, `code/`, `csv/`, and `data/v23/`).

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- **Structured JSON summary:** `bullet_cluster_audit_summary.json` (synced to root, `code/`, `csv/`, and `data/v23/`).

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3 Step 25 (V26): Cosmic Microwave Background (CMB) and Cosmological Background — Separating Numerically Integrated FLRW Background Quantities ($H(z)$, $r_s(z_*)$, $D_M(z_*)$, θ_*) from Fixed Phase/Literature Peak-Height Inputs, Negligibility of the Local $L = 10$ km Yukawa $G_{\rm eff}(a,k)$ Modification, and the Conditional Negative-ρ_Λ Separate-Branch Hypothesis

Source file: `step25_cmb_and_cosmology_en.md`

Author: Dávid Szabolcsi (ORCID: `0009-0005-6300-074X`)
Date: 2026-10-27
File Path: `scalar-tensor-yukawa-fit/step25_cmb_and_cosmology_en.md`
Previous Steps: `step1_dilaton_form_en.md` – `step24_bullet_cluster_en.md`
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Local Sources and Runnable Code:

- Runnable Python script: `scalar-tensor-yukawa-fit/cmb_and_cosmology_audit.py`
- Generated output files: `cosmological_background_models.csv`, `cmb_geff_and_acoustic_scales.csv`, `cmb_and_cosmology_summary.json` (and copies in `code/`, `csv/`, and `data/v23/`)
- Referenced theoretical steps: `step14_gravitational_lensing_null_geodesics_cosmological_background_en.md` (Section 3: homogeneous FLRW attractor), `step17_macroscopic_clock_field_closure_sparc_compatibility_en.md` (Section 4: cosmological background and limit), `step22_curved_exterior_yukawa_adm_mass_mr_and_coupled_stability` `step23_gw170817_and_lensing_en.md`, `step24_bullet_cluster_en.md`

3.1 1. Objective and V26 Methodological Delineations

In Steps 23 and 24 we established that the local finite-range ($L = 10$ km, $m_\phi c^2 \approx 1.973 \times 10^{-11}$ eV, $\alpha_Y = 1$, $\beta_{\rm Yuk} = 1/(4\pi) \approx 0.079577$) `yukawa_action_el` Einstein–Hilbert + canonical scalar branch closed in Step 22 satisfies the GW170817 $c_T = c$ bound and Solar-System PPN/Cassini constraints on the purely conformal, universally coupled Maxwell branch, while its local $L = 10$ km Yukawa fifth force is exponentially suppressed by $\exp(-r/L) \sim 10^{-10^{18}}$ on galaxy-cluster scales ($R \sim 100$–1000 kpc).

In this Step 25 (**V26**), we audit the cosmological (FLRW) background and Cosmic Microwave Background (CMB) acoustic scales via `cmb_and_cosmology_audit.py`, enforcing **three strict V26 methodological and theoretical delineations**:

1. **Explicit separation of numerically integrated FLRW background quantities from fixed analytic phase-shift and literature peak-height inputs (Review Point 2):**
   - The script `cmb_and_cosmology_audit.py` **numerically integrates** (via adaptive Gauss–Kronrod quadrature, `epsrel=1e-12`) the homogeneous FLRW Hubble function $H(z) = H_0 E(z)$, the present-day deceleration parameter q_0, the turnaround scale factor $a_{\rm max}$ ($z_{\rm turn}$) where applicable, the comoving sound horizon $r_s(z_*)$ at recombination ($z_* = 1089.80$), the comoving line-of-sight and angular distances $\chi(z_*)$ and $D_M(z_*)$, the physical angular diameter distance $D_A(z_*)$, the acoustic angle $\theta_* = r_s(z_*)/D_M(z_*)$, the acoustic multipole scale $\ell_A = \pi/\theta_*$, and the linear perturbation fifth-force modification $\Delta G_{\rm eff}(a,k)/G$ of the local $L = 10$ km Yukawa mode.
   - By contrast, the multipole columns $\ell_m^{\rm est} = \ell_A(m - \phi_m)$ (`l1_analytic_phase_est`, `l2_analytic_phase_est`, `l3_analytic_phase_est`) are **analytic phase-shift template estimates** evaluated from fixed external phase shifts (ϕ_1, ϕ_2, ϕ_3, Hu & Sugiyama 1996 / Doran & Lilley 2002), and the peak-height ratio columns (`H2_H1_lit_benchmark_input`, `H3_H2_lit_benchmark_input`) are **fixed external literature benchmark inputs** (`0.458` and `0.985` assigned to every $\Omega_c > 0$ row, and `0.312` and `0.415` assigned to every $\Omega_c = 0$ row), **not** outputs of a coupled Einstein–Boltzmann solver (CAMB/CLASS) or a spectrum fit (`boltzmann_spectrum_computed = False`).
2. **Exact scope of Planck 2018 compatibility and the input Ω_c/Ω_b ratio (Review Point 2):**

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- The ratio $\Omega_c h^2/\Omega_b h^2 = 0.1200/0.02237 = 5.364327224$ ($\Omega_c/\Omega_b = 0.264157/0.049243$) is **the input Planck 2018 CDM-to-baryon density ratio**, **not** a theoretical prediction derived from the TAL clock sector.
- Claims of "Planck 2018 compatibility" in this step are restricted **strictly** to the fact that the local $L = 10$ km Yukawa branch produces a negligible homogeneous background energy density ($\Omega_{\phi,\mathrm{can}}(z_*) \approx 9.85 \times 10^{-38}$) and a negligible linear perturbation fifth-force correction ($\Delta G_{\mathrm{eff}}/G = 2.50 \times 10^{-36}$ at the first acoustic peak $\ell_1 = 220$ and 4.65×10^{-34} in the Silk damping tail $\ell = 3000$), lying more than 31–33 orders of magnitude below the Planck 2018 sensitivity threshold and therefore leaving an input ΛCDM background undisturbed. Proving that a macroscopic clock sector can replace particle CDM across the acoustic peaks requires a closed covariant action and a full coupled Einstein–Boltzmann perturbation calculation.

3. **Conditional separate-branch status of negative $\rho_\Lambda = -5.96\times10^{-27}$ kg/m^3 and its positive $\rho_{\mathrm{clock,cosmo}} = +2|\rho_\Lambda|$ compensation (Review Point 3):**
   - In the minimal canonical action of Steps 22–23, the scalar potential is $V(\phi_L) = +\frac{1}{2}m_\phi^2\phi_L^2 \geq 0$, so a homogeneous static scalar field ($\partial_\mu\phi_L = 0$) carries a **positive** energy density in its canonical stress–energy tensor $T^{(\phi)}_{\mu\nu}$ ($\rho_{\phi,\mathrm{can}} = +V(\phi_L)/c^2 \geq 0$).
   - The negative vacuum energy density $\rho_\Lambda = -V(\phi_{\mathrm{cosmo}})/c^2 = -5.96 \times 10^{-27}$ kg/m^3 ($\Omega_\Lambda = -0.6851$) discussed in earlier manuscripts and its compensating positive cosmological clock background $\rho_{\mathrm{clock,cosmo}} = +2|\rho_\Lambda| = +1.192 \times 10^{-26}$ kg/m^3 do not follow from the minimal `yukawa_action_el` field equations of Steps 22–23. They are therefore treated strictly as a **conditional separate-branch hypothesis** (`conditional_separate_branch_hypothesis`) until an explicit action-level origin, stress–energy tensor, Bianchi identity, and initial condition are provided for both the minus sign and the compensating positive term. The compensated background ($\rho_{\mathrm{DE,eff}} = \rho_\Lambda + \rho_{\mathrm{clock,cosmo}} = +5.96 \times 10^{-27}$ kg/m^3) recovers the input ΛCDM background by algebraic construction rather than as an independent cosmological prediction.

3.2 2. Cosmological (FLRW) Behavior of the Local $L = 10$ km Yukawa Branch and Linear $G_{\mathrm{eff}}(a,k)$ Modification

3.2.1 2.1. Homogeneous Quasi-Static Attractor and Positive Canonical Energy Density

In a spatially flat or curved FLRW metric ($ds^2 = -c^2dt^2 + a^2(t)\gamma_{ij}dx^idx^j$), the unified Step 23 action $S[\phi_L]$ ($\tilde{\alpha}_L = \alpha_Y/(8\pi)$, $A = \exp(-\alpha_Y\phi_L)$) gives for the homogeneous mode $\bar{\phi}_L(t)$ the wave-operator form (both sides in m^{-2}):

$$\Big(-\frac{1}{c^2}\partial_t^2 - \frac{3H}{c^2}\partial_t - \frac{1}{L^2}\Big)\bar{\phi}_L \;=\; -\frac{\alpha_Y^2 G}{c^4}\,\bar{\varepsilon}_m \;=\; -\frac{\alpha_Y^2 G}{c^2}\,\bar{\rho}_m.$$

Here the **mass density** $\bar{\rho}_m(t)$ [kg m^{-3}] and the **energy density** $\bar{\varepsilon}_m(t) \equiv \bar{\rho}_m(t)c^2$ [J m^{-3}] are kept distinct. Multiplying by $-c^2$, the homogeneous time ODE has SI dimension s^{-2} on both sides:

$$\ddot{\bar{\phi}}_L + 3H(t)\dot{\bar{\phi}}_L + \frac{c^2}{L^2}\bar{\phi}_L \;=\; \alpha_Y^2\, G\,\bar{\rho}_m(t) \;=\; \frac{\alpha_Y^2 G}{c^2}\,\bar{\varepsilon}_m(t) \quad [\mathrm{s}^{-2}], \qquad \omega_\phi \;\equiv\; \frac{c}{L} \;=\; 2.99792458 \times 10^4\ \mathrm{s}^{-1}.$$

Dimension check: $[G\bar{\rho}_m] = (\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2})(\mathrm{kg\,m}^{-3}) = \mathrm{s}^{-2} = [\ddot{\bar{\phi}}_L] = [(c^2/L^2)\bar{\phi}_L]$ ($\bar{\phi}_L$ dimensionless). (The former right-hand side $(4\pi G\alpha_Y/c^2)\bar{\rho}_m$ had dimension m^{-2} and was wrong.) The V25 standard bookkeeping field $\bar{\phi}_{L,\mathrm{std}} \equiv \bar{\phi}_L/(4\pi)$ obeys the same equation with source $\alpha_{\mathrm{std}}G\bar{\rho}_m = (\alpha_Y/4\pi)G\bar{\rho}_m$ (at $\alpha_Y = 1$). Because the scalar mass frequency exceeds the present-day Hubble rate by 22 orders of magnitude (44 in ω_ϕ^2/H_0^2) and the recombination Hubble rate ($H(z_*) = 23\,179.39\,H_0 \approx 5.063 \times 10^{-14}$ s^{-1}) by 17 orders of magnitude:

$$\frac{\omega_\phi}{H_0} \;=\; 1.372497 \times 10^{22}, \qquad \frac{\omega_\phi^2}{H^2(z_*)} \;=\; 3.506055 \times 10^{35} \;\gg\; 1,$$

the homogeneous field adiabatically tracks the quasi-static attractor ($\gamma_L = GL^2/c^2 = 7.42616027 \times 10^{-20}$ m^3 kg^{-1}, $\bar{\rho}_m(z) = \Omega_{m,0}\,\frac{3H_0^2}{8\pi G}(1+z)^3$); both sides are identically equal:

$$\boxed{\bar{\phi}_{\mathrm{attr,unit}}(z) \;=\; \frac{\alpha_Y^2 G L^2}{c^2}\bar{\rho}_m(z) \;=\; \alpha_Y^2\gamma_L\bar{\rho}_m(z) \;=\; \frac{3}{8\pi}\,\alpha_Y^2\,\Omega_{m,0}\Big(\frac{H_0 L}{c}\Big)^2(1+z)^3,}$$

$$\boxed{\bar{\phi}_{\mathrm{attr,V25_std}}(z) \;=\; \frac{\bar{\phi}_{\mathrm{attr,unit}}}{4\pi} \;=\; \alpha_{\mathrm{std}}\gamma_L\bar{\rho}_m(z) \;=\; \frac{3}{32\pi^2}\,\alpha_Y\,\Omega_{m,0}\Big(\frac{H_0 L}{c}\Big)^2(1+z)^3 \quad (\alpha_Y = 1).}$$

Values from `cmb_and_cosmology_summary.json` (`quasi_static_attractor_epochs`, `attractor_background_model_today`) ($H_0 = 67.4$ km/s/Mpc, $\Omega_{m,0} = 0.314820$, $H_0L/c = 7.285991 \times 10^{-23}$, $\bar{\rho}_{m,0} = 2.686310 \times 10^{-27}$ kg/m^3, $\rho_{\mathrm{crit},0} = 8.532855 \times 10^{-27}$ kg/m^3, $\alpha_Y = 1$, $L = 10$ km):

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- **Today ($z = 0$):** `phi_attr_unit_kernel` = LHS $\alpha_Y^2 \gamma_L \bar{\rho}_m = \mathbf{1.994897 \times 10^{-46}}$, RHS $\frac{3}{8\pi}\Omega_{m,0}(H_0 L/c)^2(1+z)^3 = \mathbf{1.994897 \times 10^{-46}}$; `phi_attr_V25_std` = LHS $\mathbf{1.587488 \times 10^{-47}}$, RHS $\mathbf{1.587488 \times 10^{-47}}$.
  - Positive canonical energy density (standard normalization): $\rho_{\phi,\mathrm{can,std}} = \frac{c^2}{8\pi G L^2}\bar{\phi}^2_{\mathrm{attr,V25_std}} = \mathbf{1.350258 \times 10^{-76}}$ kg/m^3, $\Omega_{\phi,\mathrm{can,std}} = \rho_{\phi,\mathrm{can,std}}/\rho_{\mathrm{crit},0} = \mathbf{1.582423 \times 10^{-50}}$.
  - Positive canonical energy density (unit-kernel (V25 ϕ_L) normalization): $\rho_{\phi,\mathrm{can,unit}} = \frac{c^2}{16\pi G \tilde{\alpha}_L L^2}\bar{\phi}^2_{\mathrm{attr,unit}} = \mathbf{2.679455 \times 10^{-73}}$ kg/m^3, $\Omega_{\phi,\mathrm{can,unit}} = \mathbf{3.140162 \times 10^{-47}}$.
- **At recombination ($z_* = 1089.80$):** `phi_attr_unit_kernel` = LHS $\alpha_Y^2 \gamma_L \bar{\rho}_m = \mathbf{2.589142 \times 10^{-37}}$, RHS $\frac{3}{8\pi}\Omega_{m,0}(H_0 L/c)^2(1+z)^3 = \mathbf{2.589142 \times 10^{-37}}$; `phi_attr_V25_std` = LHS $\mathbf{2.060373 \times 10^{-38}}$, RHS $\mathbf{2.060373 \times 10^{-38}}$.
  - Positive canonical energy density (standard normalization): $\rho_{\phi,\mathrm{can,std}} = \frac{c^2}{8\pi G L^2}\bar{\phi}^2_{\mathrm{attr,V25_std}} = \mathbf{2.274509 \times 10^{-58}}$ kg/m^3, $\Omega_{\phi,\mathrm{can,std}} = \rho_{\phi,\mathrm{can,std}}/\rho_{\mathrm{crit}}(z_*) = \mathbf{4.961230 \times 10^{-41}}$.
  - Positive canonical energy density (unit-kernel (V25 ϕ_L) normalization): $\rho_{\phi,\mathrm{can,unit}} = \frac{c^2}{16\pi G \tilde{\alpha}_L L^2}\bar{\phi}^2_{\mathrm{attr,unit}} = \mathbf{4.513540 \times 10^{-55}}$ kg/m^3, $\Omega_{\phi,\mathrm{can,unit}} = \mathbf{9.845073 \times 10^{-38}}$. The physical energy density of the action $S[\phi_L]$ is the unit-kernel value; the standard value is kept for continuity with V25. Both are positive ($V \geq 0$) and negligible for the CMB.

3.2.2 2.2. Scale-Dependent Effective Gravitational Constant $G_{\mathrm{eff}}(a,k)$ and Gravitational Slip at CMB Wavenumbers

In linear cosmological perturbation theory, the effective gravitational constant governing comoving wavenumber k ($k_{\mathrm{phys}}(a) = k/a$) density perturbations on the local Yukawa branch ($\beta_{\mathrm{Yuk}} = 2\tilde{\alpha}_L \alpha_Y^2 = \alpha_Y^3/(4\pi) = 1/(4\pi) \approx 0.07957747$) is:

$$\boxed{\frac{G_{\mathrm{eff}}(a,k)}{G} = 1 + \beta_{\mathrm{Yuk}} \frac{k^2_{\mathrm{phys}}(a)\, L^2}{1 + k^2_{\mathrm{phys}}(a)\, L^2} = 1 + 2\tilde{\alpha}_L \alpha_Y^2 \frac{(kL/a)^2}{1 + (kL/a)^2} = 1 + \frac{\alpha_Y^3}{4\pi} \frac{(kL/a)^2}{1 + (kL/a)^2} \underset{kL/a \ll 1}{\approx} 1 + \frac{1}{4\pi}\Big(\frac{kL}{a}\Big)^2.}$$

$\Delta G_{\mathrm{eff}}/G = \beta_{\mathrm{Yuk}}(kL/a)^2/[1 + (kL/a)^2]$: the factor $2\tilde{\alpha}_L$ is already contained in $\beta_{\mathrm{Yuk}} = 2\tilde{\alpha}_L \alpha_Y^2 = 1/(4\pi)$ (Step 23, Section 1), so there is no extra factor of 2; this is exactly `delta_geff = BETA_YUK * k_phys_L_sq` in `cmb_and_cosmology_audit.py` (2.5015×10^{-36} at $\ell_1 = 220$, 4.6516×10^{-34} at $\ell = 3000$). The table below (`cmb_geff_and_acoustic_scales.csv`) summarizes the numerically evaluated fractional fifth-force modifications $\Delta G_{\mathrm{eff}}/G \equiv G_{\mathrm{eff}}/G - 1$, the anisotropic Weyl slip ($\eta_{\mathrm{slip}} - 1 \approx 2\,\Delta G_{\mathrm{eff}}/G$), and the late-time integrated Sachs–Wolfe (ISW) modification across representative cosmological and CMB scales:

Scale label (`scale_label`)	Multipole ℓ	k (Mpc^{-1})	$k_{\mathrm{phys}}(z_*)\,L$	$\Delta G_{\mathrm{eff}}/G(z_*)$	$\log_{10}(\Delta G_{\mathrm{eff}}/G)_{z_*}$	$\Delta G_{\mathrm{eff}}/G(z=0)$	$\log_{10}(\Delta G_{\mathrm{eff}}/G)_{z=0}$	Margin below Planck 2×10^{-2} bound (orders)
CMB_quadrupole_ISW_ell2	2.0	0.000144	5.097e-20	2.067e-40	-39.68	1.738e-46	-45.76	37.99
CMB_ISW_ell10	10.0	0.000721	2.548e-19	5.168e-39	-38.29	4.344e-45	-44.36	36.59
CMB_ISW_ell20	20.0	0.001442	5.097e-19	2.067e-38	-37.68	1.738e-44	-43.76	35.99
LSS_horizon_k98-4h	98.4	0.000067	2.383e-20	4.518e-41	-40.35	3.797e-47	-46.42	38.65
CMB_recombination_horizon	200.0	0.014420	5.097e-18	2.067e-37	-36.68	1.738e-43	-42.76	34.99
CMB_first_acoustic_peak	**220.0**	**0.015860**	**5.607e-18**	**2.502e-36**	**-35.60**	**2.102e-42**	**-41.68**	**33.90**
CMB_second_acoustic_peak	537.5	0.038753	1.370e-17	1.493e-35	-34.83	1.255e-41	-40.90	33.13
CMB_third_acoustic_peak	810.8	0.058452	2.066e-17	3.398e-35	-34.47	2.856e-41	-40.54	32.77
LSS_BAO_scale_k9.1h	934.9	0.067400	2.383e-17	4.518e-35	-34.35	3.797e-41	-40.42	32.65
CMB_Silk_damping_ell1500	1500.0	0.108138	3.823e-17	1.163e-34	-33.93	9.773e-41	-40.01	32.24
CMB_Silk_damping_tail_ell	**2000.0**	**0.144184**	**7.645e-17**	**4.652e-34**	**-33.33**	**3.909e-40**	**-39.41**	**31.63**
LSS_nonlinear_k1.0h	9341.1	0.674000	2.383e-16	4.518e-33	-32.35	3.797e-39	-38.42	30.65
LSS_small_scale_k10h	93410.1	6.740000	2.383e-15	4.518e-31	-30.35	3.797e-37	-36.42	28.65

(Note on wavenumber definitions: at the exact projection wavenumber $k_1 = \ell_1/D_M(z_*) = 0.015860$ Mpc^{-1} of the first acoustic peak $\ell_1 = 220.0$, $\Delta G_{\mathit{eff}}/G(z_*) = \mathbf{2.5015 \times 10^{-36}}$, while at the rounded reference wavenumber $k_1 = 0.016$ Mpc^{-1}, $\Delta G_{\mathit{eff}}/G(z_*) = \mathbf{2.5458 \times 10^{-36}}$; similarly, at $\ell = 3000$ ($k = 0.216277$ Mpc^{-1}), $\Delta G_{\mathit{eff}}/G(z_*) =$

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$\mathbf{4.6516 \times 10^{-34}}$, *and at rounded* $k = 0.22$ *Mpc*$^{-1}$ *it is* $\mathbf{4.8131 \times 10^{-34}}$. *Both sets of numbers are recorded in* `cmb_and_cosmology_summary.json`.)

3.3 3. Numerically Integrated FLRW Background Models vs. Fixed Phase-Shift and Literature Peak-Height Inputs

3.3.1 3.1. Equations Actually Integrated by `cmb_and_cosmology_audit.py`

For a given parameter set $(\Omega_b, \Omega_c, \Omega_\gamma, \Omega_\nu, \Omega_{\Lambda,\mathrm{eff}}, \Omega_K)$, `cmb_and_cosmology_audit.py` integrates the Friedmann equation:

$$E^2(z) \equiv \frac{H^2(z)}{H_0^2} = \Omega_{r,0}(1+z)^4 + \Omega_{m,0}(1+z)^3 + \Omega_K(1+z)^2 + \Omega_{\Lambda,\mathrm{eff}}, \qquad q_0 = \frac{\Omega_{m,0}}{2} + \Omega_{r,0} - \Omega_{\Lambda,\mathrm{eff}}.$$

The baryon–photon sound speed and the comoving sound horizon at recombination ($z_* = 1089.80$) are integrated via:

$$c_s(z) = \frac{c}{\sqrt{3\left(1 + R_b(z)\right)}}, \qquad R_b(z) = \frac{3\Omega_b}{4\Omega_\gamma(1+z)}, \qquad r_s(z_*) = \int_{z_*}^{\infty} \frac{c_s(z)\,dz}{H_0 E(z)}.$$

The comoving line-of-sight distance, curvature-corrected comoving angular distance, and acoustic scale are:

$$\chi(z_*) = \frac{c}{H_0}\int_0^{z_*} \frac{dz}{E(z)}, \qquad D_M(z_*) = \begin{cases} \dfrac{c}{H_0\sqrt{\Omega_K}} \sinh\!\left(\dfrac{\sqrt{\Omega_K}H_0}{c}\chi(z_*)\right) & (\Omega_K > 0), \\ \chi(z_*) & (\Omega_K = 0), \end{cases} \qquad \theta_* = \frac{r_s(z_*)}{D_M(z_*)}, \qquad \ell_A = \frac{\pi}{\theta_*}.$$

3.3.2 3.2. Split Summary Tables of the 8 Cosmological Models (`cosmological_background_models.csv`)

In accordance with Review Point 2, we present the 8 cosmological models in **two separate tables**: **Table 3.2.A** contains exclusively the **numerically integrated FLRW background and acoustic-scale quantities** computed by `cmb_and_cosmology_audit.py`, whereas **Table 3.2.B** reports the **analytic multipole estimates using fixed external phase shifts** (ϕ_1, ϕ_2, ϕ_3) alongside the **fixed external literature peak-height benchmark inputs** (`boltzmann_spectrum_computed = False`).

Table 3.2.A: Numerically Integrated FLRW Background and Acoustic-Scale Quantities

Model ID (`model_id`)	$\rho_{\mathrm{DE,eff}}$ (kg/m^3)	Ω_b	Ω_c	$\Omega_{\Lambda,\mathrm{eff}}$	Ω_K	Reaches $a = 1$?	q_0	$a_{\max}$ (z_{turn})	$r_s(z_*)$ (Mpc)	$\chi(z_*)$ (Mpc)	$D_M(z_*)$ (Mpc)	$100\,\theta_*$ (rad)	$\ell_A = \pi/\theta_*$
Planck18_LCDM_baseline	5.96e-27	0.04924	0.26416	0.68510	0.00000	True	−0.5276	—	144.4423	13871.12	13871.12	1.041326	301.6932
TAL_local_best_fit	5.96e-27	0.04924	0.26416	0.68510	0.00000	True	−0.5276	—	144.4423	13871.12	13871.12	1.041326	301.6932
TAL_unconstrained_negative_lambda	5.96e-27	0.04924	0.26416	−0.68510	1.37020	True	+0.8426	1.5175 (−0.3410)	144.3662	11073.14	34916.42	0.413467	759.8258
TAL_unconstrained_negative_lambda_flat	5.96e-27	0.04924	0.26416	−0.68510	0.00000	False	—	0.7718 (+0.2957)	144.4423	16367.75	16367.75	0.882484	355.9943
TAL_compensated_matter_clock_bookkeeping	5.96e-27	0.04924	0.26416	0.68510	0.00000	True	−0.5276	—	144.4423	13871.12	13871.12	1.041326	301.6932
TAL_baryon_only_diagnostic	8.10e-27	0.04924	0.00000	0.94926	0.00000	True	−0.9238	—	195.1104	27512.60	27512.60	0.709164	442.9974
TAL_baryon_only_scaling_diagnostic	8.10e-27	0.04924	0.00000	0.94926	0.00000	True	−0.9238	—	209.1445	27512.60	27512.60	0.760174	413.2712
TAL_baryon_only_open_diagnostic	0.00e+00	0.04924	0.00000	0.94926	0.00000	True	+0.0254	—	194.9249	18037.73	18631.37	1.043113	301.1911

Table 3.2.B: Analytic Phase-Shift Multipole Estimates (ϕ_m) and Fixed External Literature Peak-Height Inputs (`boltzmann_spectrum_computed = False`)

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Model ID (model_id)	Input phase shifts (ϕ_1, ϕ_2, ϕ_3)	l1_analytic_phase_est	l2_analytic_phase_est	l3_analytic_phase_est	$\Delta\ell_1^{\rm est}$ vs. Planck 220.0	H2_H1_lit_benchmark_input (fixed input)	H3_H2_lit_benchmark_input (fixed input)	boltzmann_spectru
Planck18_LCDM_reference	(0.2709, 0.2188, 0.3131)	219.95	537.38	810.62	-0.05	0.458 *(Planck/lit. $\Omega_c > 0$ input)*	0.985 *(Planck/lit. $\Omega_c > 0$ input)*	False
TAL_local_Yukawa_2710km_sector	(0.2709, 0.2188, 0.3131)	219.95	537.38	810.62	-0.05	0.458 *(Planck/lit. $\Omega_c > 0$ input)*	0.985 *(Planck/lit. $\Omega_c > 0$ input)*	False
TAL_uncompensated_negative_rhoLambda_flat	(0.2709, 0.2188, 0.3131)	558.96	1353.41	2041.58	+333.96	0.458 *(fixed $\Omega_c > 0$ input)*	0.985 *(fixed $\Omega_c > 0$ input)*	False
TAL_uncompensated_negative_rhoLambda_flat	(0.2709, 0.2188, 0.3131)	259.54	634.10	956.52	+39.54	0.458 *(fixed $\Omega_c > 0$ input)*	0.985 *(fixed $\Omega_c > 0$ input)*	False
TAL_compensated_macro_clock_background	(0.2709, 0.2188, 0.3131)	219.95	537.38	810.62	-0.05	0.458 *(Planck/lit. $\Omega_c > 0$ input)*	0.985 *(Planck/lit. $\Omega_c > 0$ input)*	False
TAL_baryon_only_flat_Lambda	(0.3150, 0.2620, 0.3550)	303.45	769.93	1171.73	+83.45	0.312 *(literature $\Omega_c = 0$ input)*	0.415 *(literature $\Omega_c = 0$ input)*	False
TAL_baryon_only_sound_diagnostic	(0.2709, 0.2188, 0.3131)	301.30 *(or 152.02 at fixed $D_M^{\Lambda CDM}$)*	736.13	1110.42	+81.30 (-67.98)	0.312 *(literature $\Omega_c = 0$ input)*	0.415 *(literature $\Omega_c = 0$ input)*	False
TAL_baryon_only_open_no_Lambda	(0.3150, 0.2620, 0.3550)	1309.70	3323.01	5057.17	+1089.70	0.312 *(literature $\Omega_c = 0$ input)*	0.415 *(literature $\Omega_c = 0$ input)*	False

3.4 4. Theoretical Assessment of Negative $\rho_\Lambda = -5.96 \times 10^{-27}$ kg/m^3 and Positive $\rho_{\rm clock,cosmo}$ Compensation (Review Point 3)

1. **Why the canonical scalar energy density is positive in the minimal action of Steps 22–23:**
   - In the Einstein-frame action specified in Step 23 (`step23_gw170817_and_lensing_en.md`, Section 2.1), the scalar potential is $V(\phi_L) = +\frac{1}{2}m_\phi^2\phi_L^2 \geq 0$. The time–time component of the standard variational stress–energy tensor $T_{\mu\nu}^{(\phi)} = \frac{c^4}{8\pi G\tilde{\alpha}_L}\left[\nabla_\mu\phi_L\nabla_\nu\phi_L - \frac{1}{2}g_{\mu\nu}((\nabla\phi_L)^2 + L^{-2}\phi_L^2)\right]$ for a spatially homogeneous field ($\nabla_i\phi_L = 0$) is:
   $$\rho_{\phi,\rm can} = \frac{1}{c^2}T_{00}^{(\phi)} = \frac{c^2}{8\pi G\tilde{\alpha}_L}\left[\frac{\dot{\phi}_L^2}{2c^2} + \frac{\phi_L^2}{2L^2}\right] \geq 0.$$
   - Consequently, the minimal canonical action of Steps 22–23 does **not** generate a negative vacuum energy density; the sign $\rho_\Lambda = -V(\phi_{\rm cosmo})/c^2 = -5.96 \times 10^{-27}$ kg/m^3 can only be introduced as a hypothesis belonging to a separate macroscopic/cosmological branch.
2. **Consequences of an uncompensated negative $\rho_\Lambda = -5.96 \times 10^{-27}$ kg/m^3 ($\Omega_\Lambda = -0.685101$) on a separate branch:**
   - **In flat space ($\Omega_K = 0$, `TAL_uncompensated_negative_rhoLambda_flat`):** today one would have $E^2(z = 0) = \Omega_{r,0} + \Omega_{m,0} + \Omega_\Lambda = -0.370202 < 0$, meaning the universe would have halted and recollapsed at $a_{\rm max} = (\Omega_{m,0}/|\Omega_\Lambda|)^{1/3} = \mathbf{0.771762}$ ($z_{\rm turn} = \mathbf{0.295737}$), never reaching $a = 1$.

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- **With spatial curvature closure ($\Omega_K = 1-\Omega_{m,0}-\Omega_\Lambda = +1.370202$, `TAL_uncompensated_negative_rhoLambda`):** the universe reaches $a = 1$ (and turns around in the future at $a_{\max} = 1.517519$), but strongly decelerates today ($q_0 = +\mathbf{0.842603}$, vs. the observed accelerating $q_0 \approx -0.53$), and the large positive curvature stretches the comoving angular distance to $D_M(z_*) = 34\,916.42$ Mpc, shifting the acoustic scale to $100\,\theta_* = \mathbf{0.413462}$ ($\ell_A = 759.83$, $\ell_1^{\rm est} \approx 553.96$).

3. **Exact status of the compensated macroscopic clock background (`TAL_compensated_macro_clock_background`):**

- If, on the conditional separate branch, one postulates a homogeneous cosmological clock condensate with energy density $\rho_{\rm clock,cosmo} = +2|\rho_\Lambda| = +1.192 \times 10^{-26}$ kg/m^3 ($\Omega_{\rm clock,cosmo} = +1.370202$, $w_{\rm clock} = -1$), the net effective dark-energy density is:

$$\rho_{\rm DE,eff} = \rho_\Lambda + \rho_{\rm clock,cosmo} = -5.96\times10^{-27} + 1.192\times10^{-26} = +5.96\times10^{-27}\ \text{kg/m}^3 \quad (\Omega_{\rm DE,eff} = +0.685101).$$

- This equality restores the input Planck 2018 ΛCDM background ($q_0 = -0.5276$, $100\,\theta_* = 1.041320$, $\ell_A = 301.6932$) **by algebraic construction**; it is **not** an independent cosmological prediction until a joint covariant action, stress–energy tensor, Bianchi identity, and dynamical attractor equation are derived for both $\rho_\Lambda < 0$ and $\rho_{\rm clock,cosmo} = +2|\rho_\Lambda|$.

3.5 5. Why the Local $L = 10$ km Yukawa Sector Does Not Replace Cosmological Cold Dark Matter (Ω_c)

If cosmological cold dark matter were omitted ($\Omega_c = 0$, retaining only baryons $\Omega_b = 0.049243$ and the local $L = 10$ km Yukawa field):

1. **Shift of the sound horizon and acoustic scale (numerically integrated background result):**

- Due to the reduced pre-recombination expansion rate $H(z)$, the sound horizon increases from $r_s(z_*) = 144.44$ Mpc to 195.11 **Mpc** (with full $N_{\rm eff} = 3.046$; or 209.14 Mpc in the semi-analytic scaling diagnostic).
- With flat Λ closure ($\Omega_\Lambda = 0.949257$), the comoving angular distance expands to $D_M(z_*) = \mathbf{27\,512.60}$ Mpc, yielding $100\,\theta_* = \mathbf{0.709167}$ and $\ell_A = \mathbf{442.9974}$ ($\ell_1^{\rm est} \approx 303.45$, vs. Planck $\ell_1 = 220.0$; or $\ell_1^{\rm est} \approx 152.02$ if evaluated at fixed ΛCDM $D_M = 13\,871.12$ Mpc). In an open dark-energy-free universe ($\Omega_K = +0.949257$), $D_M(z_*) = 118\,631.37$ Mpc, $100\,\theta_* = 0.164311$, and $\ell_A = 1911.97$.

2. **Gravitational potential wells and odd/even peak heights (literature benchmark comparison):**

- At recombination, the local Yukawa fifth-force modification is $\Delta G_{\rm eff}/G(z_*, \ell_1) = 2.50 \times 10^{-36}$, so the local $L = 10$ km field does not create baryon-free gravitational potential wells.
- Standard cosmological perturbation theory (Hu & Sugiyama 1996; Planck 2018 VI) shows that for $\Omega_c = 0$, radiation pressure erases gravitational potentials upon sound-horizon entry, suppressing the third acoustic peak ($H_3/H_2 \approx 0.415$ vs. the observed 0.985). In `cosmological_background_models.csv`, these peak-height ratios are recorded strictly as **fixed external literature benchmark inputs** (`boltzmann_spectrum_computed = False`).
- Extending the TAL macroscopic clock sector ($\rho_{\rm clock} = (c^2\Gamma_{\rm clock}/8\pi G)|\nabla\phi|^2$) to replace cosmological CDM would require deriving the input $\Omega_c/\Omega_b = 0.1200/0.02237 = 5.364327$ ratio and the evolution of acoustic potential wells from a coupled Einstein–Boltzmann system derived from a closed covariant action.

3.6 6. Cross-Step Physical Synthesis Table of Steps 22–25 (V26 — Reconciled Numerical Values)

In accordance with Review Point 4, we have corrected all three numerical discrepancies that remained in the earlier synthesis text so that every entry matches the machine-readable CSV and JSON outputs of Steps 22, 23, 24, and 25:

1. **Neutron stars (Step 22, SLy, $L = 10$ km, $\alpha_Y = 1$):** the maximum ADM mass from the exact 4-component curved exterior BVP is `M_ADM,max = 2.033214 M_sun` (`2.033213549 M_sun` at the turning point $\log_{10}\rho_{0c} = 15.316404$, with $M_{\rm surf} = 2.029106\ M_\odot$, $\Delta M_{\phi,\rm ext}^{\rm curved} = 0.004108\ M_\odot$, $M_{B,\max}^{\rm quad} = 2.423385\ M_\odot$, $R(M_{\rm ADM,max}) = 9.8853$ km, $R_{1.4}^{\rm ADM} = 11.5359$ km).

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2. **GW170817 sub-cutoff scalar amplitude suppression (Step 23, $D = 40$ Mpc, $f = 100$ Hz $\ll f_c = 4.7713$ kHz):** the base-10 logarithm of the evanescent suppression factor is `log10 |delta phi / delta phi_0| = -5.360e19` (`-5.360058e+19` at $f = 100$ Hz and `-5.360179e+19` in the static $f \to 0$ limit in `gw170817_dispersion_and_cT.csv`).
3. **Bullet Cluster synthetic 1D convergence test (Step 24):** the normalized synthetic 1D target MSE of the local $L = 10$ km Yukawa branch is `M_synth = 32.089` (`synth_mse_norm_vs_target = 32.088829` in `bullet_cluster_peak_offsets_summary.csv`) and is explicitly labeled as the synthetic 1D target MSE ($M_{\rm synth}$), not an observational $\chi^2_{\rm red}$.

Physical System / Test	Characteristic Scale r or k^{-1}	r/L or $(k_{\rm phys}L)^{-1}$ ($L = 10$ km)	Computed Result of the Local $L = 10$ km Yukawa Branch (`yukawa_action_el`)	Observational Constraint / Status (V26)	Is a Separate Macroscopic / Cosmological Branch Required?
Neutron Stars (TOV + Curved Exterior ADM + Stability, Step 22)	$R_* \approx$ 9.89–13.9 km	$r/L \sim 1.0$	SLy ($L =$ 10 km, $\alpha_Y = 1$): `M_ADM,max = 2.033214 M_sun` ($M_{\rm surf} =$ $2.029106\,M_\odot$, $\Delta M_{\phi,\rm ext} =$ $0.004108\,M_\odot$, $M^{\rm quad}_{B,\rm max} =$ $2.423385\,M_\odot$, $R(M_{\rm ADM,max}) =$ 9.8853 km, $R^{\rm ADM}_{1.4} =$ 11.5359 km); dynamical $\omega_0^2 = 0$ zero crossing within $+5.80 \times 10^{-6}$ dex of the ADM peak	Satisfies PSR J0740+6620 and NICER mass–radius bounds (global stability statement remains conditional pending independent free-surface variation closure)	**No** — the local $L = 10$ km Yukawa branch self-consistently describes compact-star structure
GW170817 / GRB 170817A Propagation and Polarization (Step 23)	$D = 40$ Mpc $=$ 1.234×10^{21} km	$D/L =$ 1.234×10^{20}	On the purely conformal, universally coupled Maxwell branch: $c_T = c$, $\Delta c_T/c = 0$, $\Delta t_{g-\gamma} = 0$ s; $f_c = 4771.345$ Hz; at $f = 100$ Hz, evanescent scalar suppression: `log10|dphi/dphi0| = -5.360e19`; at $f = 5$ kHz, $\Delta t_\phi =$ $+1.4645 \times 10^{15}$ s	Satisfies the Abbott et al. (2017) $\lvert\Delta c_T/c\rvert \le$ 3×10^{-15} bound on the purely conformal branch	**No** (though $c_T = c$ must be proven separately on any disformal $D(\phi_L, X) \neq 0$ branch)

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Physical System / Test	Characteristic Scale r or k^{-1}	r/L or $(k_{\rm phys}L)^{-1}$ ($L = 10$ km)	Computed Result of the Local $L = 10$ km Yukawa Branch (`yukawa_action_el`)	Observational Constraint / Status (V26)	Is a Separate Macroscopic / Cosmological Branch Required?
Solar–Cassini PPN and Light Deflection ($1.000000\,M_\odot$, Step 23)	$b_{\rm Cassini} = 1.6\,R_\odot = 1.11312 \times 10^6$ km	$b/L = 1.11312 \times 10^5$	Dedicated $1.0\,M_\odot$ Solar model: $\hat{\alpha}_{\rm GR,\odot}(R_\odot) = 1.751190''$, $\hat{\alpha}_{\rm GR,\odot}(1.6\,R_\odot) = 1.094494''$; $\lvert\gamma_{\rm PPN,\odot}(1.6\,R_\odot) - 1\rvert = \mathbf{10^{-48\,343.00}}$ ($\beta_{\rm Yuk} = 1/(4\pi)$); conformal photon invariance yields exact null-geodesic deflection $\delta\hat{\alpha}_{\rm null} = 0$	$> 48\,338$ orders of magnitude below the Cassini $\lvert\gamma_{\rm PPN} - 1\rvert \le 2.3 \times 10^{-5}$ bound	**No** — the local Yukawa branch is naturally screened at $r \gg 10$ km
SPARC Galactic Rotation Curves (Steps 13, 17, 19)	$r \sim$ 1–50 kpc	$r/L \sim 3\times10^{15}$–1.5×10^{17}	The local $L = 10$ km Yukawa force vanishes as $\exp(-r/L) \sim 10^{-10^{15}}$, reverting to pure baryonic Newtonian gravity	Reproducing SPARC RAR / BTFR requires macroscopic coupling	**Yes** — macroscopic clock-field closure ($\rho_{\rm clock} \propto \lvert\nabla\phi\rvert^2$) or effective a_0
Bullet Cluster (1E 0657-558) Weak Lensing (Step 24)	$R \sim$ 100–1000 kpc	$R/L \sim 3 \times 10^{17}$–$3 \times 10^{18}$	$\exp(-100\text{ kpc}/10\text{ km}) = 10^{-1.340\times10^{17}}$; κ peaks remain on the X-ray gas ($\Delta x = 0$ kpc); synthetic 1D target MSE: `M_synth = 32.089` (baryonic GR: `32.089`, gas-dominated $L_{\rm macro} =$ 300 kpc: `20.723`, collisionless galaxy-sourced clock field: `1.637`)	The $\sim$ 180–200 kpc lensing–X-ray offset cannot be produced by a local gas-sourced field	**Yes** — requires particle CDM or a galaxy-specific ($\eta_{\rm gal}/\eta_{\rm gas} \ge 25.665$) nonlocal clock branch

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Physical System / Test	Characteristic Scale r or k^{-1}	r/L or $(k_{\rm phys}L)^{-1}$ ($L = 10$ km)	Computed Result of the Local $L = 10$ km Yukawa Branch (`yukawa_action_el`)	Observational Constraint / Status (V26)	Is a Separate Macroscopic / Cosmological Branch Required?
Cosmological FLRW Background and CMB Acoustic Scale (Step 25)	$k^{-1} \sim 5$–10^4 Mpc	$(k_{\rm phys}L)^{-1} \sim 10^{16}$–$10^{20}$	$\Omega_{\phi,\rm can}(z_*) = 9.85 \times 10^{-38} \geq 0$; $\Delta G_{\rm eff}/G(z_*, \ell_1 = 220) = \mathbf{2.50 \times 10^{-36}}$, $\Delta G_{\rm eff}/G(z_*, \ell = 3000) = \mathbf{4.65 \times 10^{-34}}$; local Yukawa sector leaves input ΛCDM background unchanged ($100\,\theta_* = 1.041320$, $\ell_A = 301.6932$)	Local Yukawa modification is > 31–33 orders below Planck 2018 $G_{\rm eff}$ bounds; $\Omega_c/\Omega_b = 5.364327$ is an input Planck ratio	**Yes**, if CDM (Ω_c) or dark energy (Ω_Λ) is to be derived from the clock sector (requires coupled Boltzmann calculation)

3.7 7. Checks and Reproducibility

All calculations in Step 25 can be reproduced with a single command:

```
python3 cmb_and_cosmology_audit.py
```

Running the script generates and verifies the following files (in the root directory and synced to `csv/` and `data/v23/`):

- `cosmological_background_models.csv`: numerically integrated FLRW background quantities ($E^2(z=0)$, $E(z_*)$, q_0, $a_{\rm max}$, $r_s(z_*)$, $\chi(z_*)$, $D_M(z_*)$, $D_A(z_*)$, $100\,\theta_*$, ℓ_A) for all 8 models, together with dedicated columns for fixed phase-shift estimates (`l1_analytic_phase_est` – `l3_analytic_phase_est`) and fixed literature peak-height benchmark inputs (`H2_H1_lit_benchmark_input`, `H3_H2_lit_benchmark_input`, `boltzmann_spectrum_computed = False`).
- `cmb_geff_and_acoustic_scales.csv`: $\Delta G_{\rm eff}(a,k)/G$, Weyl slip, and ISW modifications across 13 cosmological and CMB wavenumbers ($\ell = 2$ through $k = 10\,h$ Mpc^{-1}).
- `cmb_and_cosmology_summary.json`: machine-readable audit protocol recording the V26 theoretical disclosures (`computed_vs_external_inputs_disclosure`, `conditional_separate_branch_hypothesis`), background integrals, and reconciled synthesis numbers across Steps 22–25 (`v26_synthesis_table_alignment`).

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